PREVENTION OF NOSOCOMIAL INFECTION

Unit of learning code- HE/CU/TT/CR/02/6/A

Related Unit of Competency in Occupational Standard: Prevent Nosocomial Infections
Introduction to the unit of learning

This unit describes the competencies required to Prevent Nosocomial Infections.

This unit describes the competencies required to prevent and control nosocomial infections. It involves handling Operation Theatre instruments and patients in the Operation Theatre room. Necessary precautions are taken in the context of theatre setting to avoid infections.

Summary of Learning Outcomes

1. Prepare surgical instruments
2. Prepare patient for surgery
3. Dispose theatre waste

Learning Outcome 1 Prepare surgical instruments

Introduction to the learning outcome

This learning outcome specifies the content of competencies required to plan for Prevent Nosocomial Infections. It includes:

- Infection prevention
- Personal OT protective gears
- Sterilization
- Decontamination
- Waste management
- Aseptic techniques
- Types of surgical instruments
- Types of surgical packs and sets
- Types of surgery
- Sterilization in Operation Theatre
- Safety practices in Operation Theatre
- Packing of surgical instruments
- Theatre layout
- Laminar flow in Operation Theatre

Performance Standard

1. Instruments and packs are decontaminated using detergents according to manufacturer's instructions, laid down policies and WHO standards
2. Instruments and packs are packed according to surgical procedure manual
3. Instruments and packs are sterilized according to WHO standards

4. Surgical packs and instruments are selected according to the surgical procedure to be done and WHO
5. Laminar flow system is operated as per the recommended manufacturers' instruction

1.2.1.3 Information Sheet

Infection prevention

Infection prevention is a scientific approach and practical solution designed to prevent harm caused by infection to patients and health workers.

Personal OT protective gears

These are physical barriers used to prevent transmission of infection to the patient or from the patient to the persons attending them. They include Gloves, Gowns/Aprons, Surgical masks, Protective eyewear, Headgear and Boots.

Gowns and Aprons

These should be made of waterproof material and worn full size to give maximum protection.

They should be used when splashing of blood or other body fluids or when any other potentially infectious materials are anticipated.

Gloves

Provide a barrier against potentially infectious microorganisms in blood, other body fluids, and medical waste, thus lowering the risk of transmitting infections to both health care workers and patients. Sterile surgical gloves should be worn during all procedures in which the main aim is to avoid introduction of pathogens into the patient, for example during surgical procedures. Single use examination gloves should be worn for all procedures in which the healthcare worker will be in contact with intact mucous membranes.

Masks and Eye Shields

These provide some protection against airborne pathogens and shield against splashes.

Headgear and Boots

Caps are worn full length to cover the head. Boots and shoe covers should be waterproof


Personal Protective Equipment (PPE)
Respirators and Masks
Medical Gowns, Aprons, Caps and Shoes
Glasses, Goggles and Face Shields
Medical Gloves


Hair cover
Mask
Gown
Gloves
Leg covers
Booties

Sterilization

Sterilization Processes by which all pathogenic and non-pathogenic microorganisms, including endospores, are killed. This term refers only to a validated process capable of destroying all forms of microbial life, including endospores. The sterilizer is a piece of equipment used to attain either physical or chemical sterilization. The agent used must be capable of killing all forms of microorganisms.

Decontamination

Cleaning reusable items with an approved sporicidal disinfectant to render the item safe for handling. Decontamination of instrumentation is performed in a designated area by specially trained personnel immediately after completion of the surgical procedure. The scrub person can facilitate the instrument decontamination process by wiping instruments as they are used on the sterile field and then opening the instruments completely before placing in a tray for return to the processing area. This is referred to as point of use cleaning. An enzymatic cleaner such as a foam or solution can applied to the instruments to prevent debris from drying during transport to the central service area. All instruments on the table during a surgical procedure require decontamination before processing to the required level of safety for patient use.

Decontamination combines mechanical or manual cleaning and a physical or chemical microbicidal process. Prerinsing, washing, rinsing, and disinfecting/sterilizing is done in the processing department to render the instrumentation safe for handling. This part of the process is referred to as thorough cleaning or return-to-use cleaning.

Waste management

Health care waste is a potential reservoir of pathogenic microorganisms and requires appropriate, safe, and reliable handling. Safe management of health care waste is a key issue in controlling and reducing HAIs. There should be a person or persons responsible for the organization and management (collection, storage, and disposal) of waste. Waste management should be conducted in coordination with the infection-control team.

Purpose of proper waste management

- Protect people who handle waste items from accidental injury
- Prevent the spread of infection to patients, clients, and HCWs
- Prevent the spread of infection to the local community
- Safely dispose of hazardous materials

Categorization of health care waste

1. Infectious waste (e.g., tissues, materials, or equipment that have been in contact with blood or other body fluids).

2. Pathological waste (e.g., tissues, organs, body parts, blood, infected animals from laboratories, and body fluids).
3. Sharps (e.g., needles, hypodermic needles, scalpels, and other blades, knives, infusion sets, saws, and broken glass)
4. Pharmaceutical waste (e.g., expired, spilt, and contaminated pharmaceutical products, discarded bottles or boxes with residues, and drug vials).
5. Radioactive waste includes solid, liquid, and gaseous materials contaminated with radionuclide.
6. Genotoxic/cytotoxic waste may include certain cytotoxic drugs often used in cancer therapy, vomit, urine or feces from patients treated with cytotoxic drugs, chemicals, and radioactive material.
7. Chemical waste consists of discarded solid, liquid, and gaseous chemicals (e.g., from diagnostic and experimental work and from cleaning, housekeeping, and disinfecting procedures).
8. Waste with heavy metal content includes waste containing mercury, cadmium, lead, and drugs containing arsenic, among others.
9. Non-infectious/general waste includes waste generated from offices, kitchens, packaging material, and from stores. It is similar to domestic waste.
10. Other wastes generated from health care settings include:

- Electronic waste.
- Construction waste.
- Obsolete equipment/furniture.

Principles of Waste Management

Key steps in the management of health care waste are as follows:

1. Waste minimization

2. Segregation (separation)

3. Collection

4. Transportation

5. Storage

6. Treatment

7. Disposal

Summary of key steps in health care waste management

Minimization: Refers to approaches adopted by the health facility to reduce the amount of HCW generated during delivery of services. It includes strategies to reduce unnecessary injections, as well as to recycle or reuse some of the materials.

Segregation: Refers to placing HCW into separate containers according to type: non-infectious or general waste, infectious, highly infectious, and sharps waste.

Handling and storage: Refers to steps taken to manage waste during containment and storage whilst waiting for collection or transportation to a treatment plant or disposal site.

Collection and transport: Refers to an organized system for removing waste from the point of generation or temporary storage to a treatment or disposal site. Waste may be transported within the health facility or to an offsite treatment plant and disposal site.

Treatment: Refers to rendering HCW safe for handling and final disposal. Some of the methods used include:

- Incineration: burning at high temperatures in an incinerator—850°C to 1100°C (Demontfort 600°C to 700°C).
- Sterilisation using autoclave or microwave technology.
- Chemical disinfection: treatment methods using a chemical such as hypochlorite solution (jik) to render the waste safe.
- Shredding waste using mechanical grinders to break it down into unrecognizable pieces. This method does not treat infectious waste and should be used in conjunction with sterilization.
- Macerators for anatomical waste fall in this category of treatment.

Figure 1. Key steps for health care waste management

Disposal: Refers to the final discharge of waste and residues or by-product from the treatment of waste. Some of the common methods of disposal are:


General waste
Paper
Packaging material
Food
General waste
Infectious waste
Gauze/dressing
Used IV/ fluid lines
Used gloves
Infusion set
Infectious waste
Pathological waste
Anatomical waste
- Teeth
- Placenta
Pathological waste
- Sputum container
- Test tube containing specimen
Pathological waste
Sharp Waste
Cannula/branula
Broken slides
Broken vial
Broken ampules
Lancet
Retractables
Scalpels
Blades
Needles
Suture needles
SAFETY BOX

Aseptic techniques

The methods by which microbial contamination is contained in the environment are referred to as aseptic technique.

Some key elements of asepsis include the following :

Items have been cleaned and decontaminated so they are safe to handle with clean, bare hands.

Items in use in patient care are handled with examination gloves for the protection of both the caregiver and the patient.

Items have been cleaned, decontaminated, disinfected, or terminally sterilized without a wrapper, and stored in a clean, dry place.

Items may start out sterile but are not maintained or used under sterile conditions. Skin preps may be packaged sterile, but skin cannot be sterilized. The process is aseptic.

Contamination is contained. Extraneous contamination is avoided.

Items are set up on clean towels or drapes and used with examination gloves.

Items are not sterile or maintained sterile during use. Extraneous contamination is avoided.

Disposable items are not cleaned and reused for another patient.

Items are classified as semicritical or noncritical by Spaulding's classification of the importance of patient care items.

Items can be used outside the confines of the restricted area.

Items are used on intact skin or mucous membranes.

Items are not used when the patient's vascular system will be entered.

Types of surgical instruments

Classes of instrument by function

Cutting instruments and dissecting

Grasping or holding instruments

Hemostatic instruments.

Retractors

Tissue unifying instruments and materials: needle holders, surgical needles, staplers, clips

Instruments used for dissecting and cutting

They have sharp edges. They are used to dissect, incise, separate or excise tissue. They should be kept separate from other instruments. Examples:

Scalpels, knives, scissors

Types of Scissors

Blades may be straight, angled or curved.

It may also be pointed or blunt at the tip

To maintain the sharpness scissors should be used for intended purpose

Sutures scissors-Used to cut sutures and supplies. have blunt points to avoid cutting structures near the suture

Tissue /dissecting scissors-Most have sharp blades .curved or angled blade are needed

mayo scissors- they are curved used to cut heavy tissues

Metzbaum(metz)- used to cut delicate tissues

Wire scissors- have short heavy blades. They are used to cut stainless steel sutures

Dressing/bandage scissors- used to cut drains, dressing.

Bone cutters/ debulking tools-They are suitable for cutting into or through a bone and cartilages. They include chisel, osteomes, gauges, rasps and files

Curette- tissue / bone is removed by scrapping with sharp edge of the loop

Grasping and holding

Tissues should be grasped and held in a position so that the surgeon can perform the desired manoeuvre without injuring the surrounding tissue.

Dissecting forceps

Smooth/thumb forceps- resemble tweezers.They have serrations at the tip used for grasping delicate tissues

Toothed forceps- have single tooth on one side that fits perfectly between the teeth on the opposite side. Provide firm hold on tough tissue i.e skin

Allis forceps- each jaw curves slightly inward and there is raw of teeth at the end

Babecock- the end of jaw is rounded section and fenestrated. Used in grasping delicate tissue

cocker

Tenaculums- curved or angled points at the end of jaws. Penetrate tissues to grasp firmly

Stone forceps- used to grasp calculi

Bone holders used to stabilise the bone

Clamping and occluding

They are used to apply pressure. Mostly used for clamping blood vessels

Haemostats/artery forceps- have either straight or curved jaws. Have serrations that goes across the jaws.

Crushing clamps- used to crash or clamp blood vessels

Non-crashing clamps- used to occlude major blood vessels temporarily which minimizes tissue trauma

Exposing and retracting

They hold organs and tissues away from surgical site so that the surgeon can see clearly and Can have room to operate EXAMPLES

Handled retractors- they have a blade on handle. They are used in pairs and are used by first and second assistant i.e malleable- may bent to desired angle or length

Hooks- they are used to retract delicate tissues

Self retaining retractors- holding device with two or more blades. Can be inserted to spread the edges of an incision and hold them apart.

Viewing

Used for viewing interior body cavities, hollow organs and structures and include speculum and endoscopies

Suctioning and aspirating

Blood, body fluids and irrigating solutions may be removed by mechanical suction or manual aspiration

Types of surgical packs and sets

Classification of sets

Specific sets

General sets

Inspect each instrument before and after use to detect imperfections.

Set aside damaged instruments and send them for repair or replacement promptly.

Use instruments only for the purpose for which they are designed.

Handle instruments gently

Provide protection for delicate instruments

Clean instruments meticulously.

Keep oil away from instruments

Give instruments regular maintenance.

Take inventory

Rotate sterile instruments in order to use item according to expiring date

Sterile packs may be stored up to the expiry date

Types of surgery

Surgery is defined as the art of science of treating diseases, injuries and deformities by operations and instrumentations

Classification of surgery

Elective-Does not require urgent intervention, it is planned and performed when patient chooses.

Urgent -Necessary for patient's health. Is indicated to prevent additional health problems from occurring

Emergency-Performed immediately to save the individual's life or preserve function of a body part

Required-Patient needs to have surgery, Plan within a few weeks or month i.e Prostatic hyperplasia without bladder obstruction, Cataracts

Optional-Decision rests with patient and it is a Personal preference i.e Cosmetic surgery

<u>Sterilization in Operation Theatre</u>

Bacterial endospores (e.g., clostridia and bacillus) are the most resistant of all living organisms because of their capacity to withstand external destructive agents. Although the physical or chemical process by which all pathogenic and non-pathogenic microorganisms (including endospores) are destroyed is not absolute, supplies and equipment are considered sterile when all

parameters have been met during a sterilization process.

Reliability Parameters for Sterilization

Two types of parameters are considered for the reliability of sterilizing methods: product-associated parameters and process-associated parameters.

Product-Associated Parameters

Bioburden. The degree of contamination with microorganisms and organic debris

Bioresistance. Factors such as heat and/or moisture sensitivities and product stability

Biostate. The nutritional, physical, and/or reproductive phase of microorganisms

Bioshielding. Characteristics of the packaging materials

Density. Factors affecting penetration and evacuation of the agent

Process-Associated Parameters

Temperature

Humidity/moisture/hydration

Time

Purity of the agent and air, and the residual effects or residues

Saturation/penetration

Capacity of the sterilizer and the position of items within the chamber

Methods of Sterilization

Reliable sterilization depends on the contact of the sterilizing agent with all surfaces of the item to be sterilized. Selection of the agent used to achieve sterility depends primarily on the nature of

the item to be sterilized. The time required to kill endospores in the available equipment then becomes critical. Sterilization processes are either physical or chemical, and each method has its

advantages and disadvantages. The following are available sterilizing agents (sterilants):

1. Thermal (physical)

a. Steam under pressure/moist heat

b. Hot air/dry heat

2. Chemical

a. Ethylene oxide gas

b. Formaldehyde gas and solution

c. Hydrogen peroxide plasma/vapor

d. Ozone gas

e. Acetic acid solution

f. Glutaraldehyde solution

g. Peracetic acid 0.2% solution

h. Hypochlorous acid (electrochemical conversion process)

3. Radiation (physical)

a. Microwave (nonionizing)

b. X-ray (ionizing)

Sterilization Cycle

The time required to achieve sterilization is referred to as the process cycle, which includes the following:

• Heat up and/or penetration of the agent

• Kill time (i.e., exposure to the agent)

• Safety factor for bioburden

• Evacuation or dissipation of the agent

Chemical Sterilizing Agents

The chemical agents used to sterilize heat-sensitive items can be toxic or can vaporize to emit noxious fumes that are irritating to the eyes and nasal passages, even at low levels of exposure.

Ethylene Oxide

Ethylene oxide (EO) is used in a gaseous form for sterilization and is a known mutagen and carcinogen. The residual products can be toxic if there is direct contact with the skin or the gas is inhaled. Exposure can cause dizziness, nausea, and vomiting. Ethylene glycol and ethylene chlorohydrin are by products of a reaction with moisture, such as on the hands. All porous items sterilized with EO should be aerated to dissipate the gas.

Formaldehyde

Formaldehyde may be used in a gaseous or liquid form. The vapors are toxic to the respiratory tract. Formaldehyde is a potent allergen, mutagen, and carcinogen, and it can cause liver toxicity.

Glutaraldehyde

Glutaraldehyde is the least toxic of the three sterilizing agents, but the fumes from the liquid form may be irritating to the eyes, nose, and throat. Contact dermatitis and hives have been reported. Glutaraldehyde should be used only in a closed container and in a well-ventilated area.

Thermal Sterilization

Heat is a dependable physical agent for the destruction of all forms of microbial life, including endospores. It may be used moist or dry. The most reliable and commonly used method of sterilization is steam under pressure.

Steam Under Pressure (Moist Heat Sterilization)

Heat destroys microorganisms, and this process is hastened by the addition of moisture. Steam in itself is inadequate for sterilization. Pressure that is greater than atmospheric pressure is necessary to increase the temperature of steam for the thermal destruction of microbial life. Moist heat in the form of steam under pressure causes the denaturation and coagulation of protein or the enzyme protein system within cells. Direct saturated steam contact is the basis of the steam sterilization process. For a specified time and at a required temperature, the steam must penetrate every fiber and reach every surface of the items to be sterilized. When steam enters the sterilizer chamber under pressure, it condenses on contact with cold items. This condensation liberates heat, simultaneously heating and wetting all items in the load and thereby providing the two requisites:

moisture and heat. This sterilization process is spoken of in terms of degrees of temperature and time of exposure—not in terms of pounds of pressure. Pressure increases the boiling temperature of water but in itself has no significant effect on microorganisms or steam penetration.

Exposure time depends on the size and contents of the load and the temperature within the sterilizer. At the end of the cycle, reevaporation of water condensate must effectively dry contents of the load to maintain sterility; the water is dried from the sterilized pack or item.

The vegetative forms of most microorganisms are killed in a few minutes at temperatures ranging from 130° F to 150° F (54° C-65° C); however, bacterial endospores will withstand a temperature of 240° F (115° C) for more than 3 hours. No living thing can survive direct exposure to saturated steam at 250° F (121° C) for longer than 15 minutes. As the temperature of the steam is increased, the time of exposure may be decreased. A minimum temperature-time relationship is maintained throughout all portions of the load to accomplish effective sterilization.

Advantages of Steam Sterilization

Steam sterilization is the easiest, safest, and surest method of onsite sterilization. Heat- and moisture-stable items that can be steam sterilized without damage should be processed with this method.

Steam is the fastest method; its total time cycle is the shortest.

Steam is the least expensive and most easily supplied agent. It is piped in from the facility's boiler room. An automatic, electrically powered steam generator can be mounted beneath the sterilizer for emergency standby when steam pressure is low.

Most sterilizers have automatic controls and recording devices that eliminate the human factor from the sterilization process as much as possible when operated and cared for according to the recommendations of the manufacturer.

Steam leaves no harmful residue. Many items such as stainless steel instruments withstand repeated processing without damage.

Disadvantages of Steam Sterilization

Precautions must be use used in preparing and packaging items, loading and operating the sterilizer, and drying the load.

Items need to be clean, free of grease and oil, and not sensitive to heat.

Steam must have direct contact with all areas of an item. It must be able to penetrate packaging material, but the material must be able to maintain sterility.

The timing of the cycle is adjusted for differences in materials and sizes of loads; these variables are subject to human error.

Steam may not be pure. Steam purity refers to the amount of solid, liquid, or vapor contamination in steam. Impurities can cause wet or stained packs and stained instruments.

Dry Heat Sterilization

Dry heat in the form of hot air is used primarily to sterilize anhydrous oils, petroleum products, and talc, which steam and ethylene oxide gas cannot penetrate. The destruction of microbial life

by dry heat is a physical oxidation or slow burning process that involves coagulating the protein in cells. Higher temperatures are required in the absence of moisture because the microorganisms

are destroyed through a very slow process of heat absorption by conduction. Dry heat sterilizers are not commonly found in the OR processing areas. Dry heat is commonly used in dental suites.

Advantages of Dry Heat Sterilization

Hot air penetrates certain substances that cannot be sterilized by steam sterilization or another method.

Dry heat is a protective method of sterilizing some delicate, sharp, or cutting-edge instruments. Steam may erode or corrode cutting edges.

Disadvantages of Dry Heat Sterilization

A long exposure period is required, because hot air penetrates slowly and possibly unevenly.

The time and temperature required will vary for different substances.

Overexposure may ruin some substances.

Safety practices in Operation Theatre

Surgical care is a fundamental part of healthcare around the globe. Ensuring surgical safety in operation theatres is the need of the hour. A safe operating theatre is achieved only through careful planning, maintenance and periodic checks, as well as proper ongoing training for staff. An operating theatre is a complex system with numerous risk factors, including not only the features of the structure and its fixtures, but also the management and behaviour of healthcare workers. Safe surgeries ensure less morbidity and mortality, improved quality of work, less liability, and greater efficiency.

There is increasing evidence that to achieve the full benefit of the checklist, there needs to be an understanding of, and a strategy for, mitigating the technical, socio-political, and psychological barriers to using the checklist. Freestanding surgical units may need to be particularly vigilant in ensuring that personnel and equipment are in good condition for surgery. Protocols and procedures to identify and manage stress and fatigue in surgical personnel may help to avoid surgical errors and patient injuries. The operating room is an appropriate educational environment, but the presence of observers at any level must not be allowed to compromise patient safety. Patient safety in surgery demands skilled individuals using well-functioning equipment under adequate supervision.

Modern Issues of Patient Safety

Medical errors are inescapable in healthcare profession. The identification of causes and implementation of safety plans to reduce these errors will help to establish an effective patient safety in operation theatres. The causes advocated by WHO are enlisted below:

• Lack of continuous training and education

• Past tolerance of unsafe practice

• Lack of regulations /rules

• Gaps in communication among different healthcare providers

• Gaps in communications between healthcare providers and patients

• Unstable /unreliable systems

• Fear of admission of guilt / wrong doing

• Human factors

Special Considerations for Surgical Procedures

Procedure Scheduling

Scheduling the procedure is very important and is a form of OT planning. All surgery requests should be made in writing and verbal requests by the medical staff should be avoided to prevent any errors in scheduling the procedure. Verification of every document including consent, history, and surgeon orders at the time of scheduling is necessary. Electronic medical records improve the safety process, decreases misunderstandings and missing documents.

Pre-operative measures to prevent errors

A preoperative check up by the concerned doctor gives an added opportunity to correct any mistakes and inconsistencies in the documentation. All discrepancies should be mended at this step before proceeding further with surgical procedure. The informed consent should be duly signed before the procedure. Patient should be explained about the surgical procedure, associated complications, additional procedures, placement of foreign material (Meshes, Plates, Screws, Stents). The next approach is to mark the site of the procedure to avoid wrong site surgery. The formulated guidelines for marking the site are; involving the patient while performing site marking, to be marked by a licensed practitioner who is responsible for the procedure and the marks should be unambiguous and uniform within the institution and should be semi-permanent to be visible after skin preparation and draping.

The institution should have a written process to ensure that the correct site is operated on. Alternatively, radiopaque markers can be used in the procedures involving fluoroscopy. Surgical armamentarium (e.g. required instruments, guide wires, laser fibres, scopes, stents, loops, prosthesis, etc.) should be verified prior to procedure.

In OT Just Before the Commencement of Surgery

Implementation of safety checklist as proposed by WHO is the main objective at this stage. This helps in improving the outcomes. The timeout is a critical advantage to pause and be sure that the team is aligned about what is being done to whom in which location," Double- and triple-checking a patient's surgical information is worth the extra time to avoid errors. Conduction of timeouts on electronic white boards ensures correct patient identification, site marking, draping and other essential elements are followed. The collaboration of team members in this task is essential at every step for smooth work up. In the modern age of medicine, digital images, displaying the CT-scan, X-rays on an auxiliary monitor during the procedure act as a guide for surgeons and improve patient safety.

The consequences of positioning related injuries are preventable but can be profound and can result in morbidity, and litigation. Neurological, vascular, musculoskeletal, and pressure ulcers are the most common position related injuries in surgical patients. Neurological complications can be avoided by placing forearms in neutral position or slightly supinated to minimise pressure in the cubital tunnel. It is advised to strap and place them adequately to maintain the correct limb position during the procedure even if the surgical table is moved. The patient’s head should be placed in a neutral position and the arm should not exceed abduction of more than 90° to avoid brachial plexus injury. Straps should not be too tight to prevent ischemia and compartmental syndrome. Padding under osseous prominences can help avoid pressure-related complications. Surgeons must be observant to avoid possible compartment syndrome (limbs) when positioning patients for open, endoscopic, and laparoscopic surgeries.

Collection of Biopsies and Surgical Specimens

A large amount of patients, multiple samples from the same patient, lack of staff, and lack of continuous education and training of healthcare workers account for medical errors. Documentation, writing policies and protocols detailing responsibilities makes the task simpler and avoid errors. To make the process as simple as possible, reducing the number of steps between collecting the samples and receiving the laboratory report should be encouraged.

Procedures that involve biopsy and tissue sampling a specimen may pass through the hands of more than twenty individuals in several workplaces until the final pathology report is given. These handoffs significantly increase the risk of a mix-up and can lead to serious diagnostic errors. Mutual cooperation for supervision of clinicians, technicians, and administrative assistants is essential to prevent and detect errors. The most precarious steps of the biopsy process include labelling of the specimens, appropriate request forms, and accessioning of biopsy specimens. The modern methods for data entry, automated systems for patient identification and specimen labeling, as well as two or more identifiers during sample collection are important steps to reduce misidentification. If misidentification is detected, rejection then recollection is the most suitable approach to manage the specimen.

The introduction of innovative methods DNA analysis to assist with correct identification can be used when recollection is not available.

Postoperative Discharge Work up

Planning a discharge has been shown to influence patient outcomes; patient safety can prevent readmissions and improve patient satisfaction. Patient education (Postoperative instruction charts) is when they are discharged home with catheters, stomas, stents, drains, or any other medical device that needs special care. Patient education can reduce complications and improve their quality of life after surgical procedures. Healthcare workers must be aware that language barriers, socioeconomic status, and age can impact patient comprehension of instructions.

Medication Safety During Administration

Medication safety is ensured by applying the five R's: right drug, right route, right time, right dose, and right patient. Medication errors are barriers that prevent the right patient from receiving the right drug in the right dose at the right time through the right route of administration at any stage during medication use, with or without the occurrence of adverse drug events.

Modern systems of information technology, such as computerised order entry, automated dispensing, barcode medication administration, electronic medication reconciliation, and personal health records are essential in the prevention of medication errors. Electronic medical records help in rapidly screening the medication regimens of hospitalised patients and deliver timely, point-of-care intervention when indicated.

The most common prescribing errors are incorrect drug, incorrect dose, allergies, and drug-drug interaction. It is important to tailor prescriptions for individual patients, identifying allergies, pregnancy, lactation, age, co-morbidities. Healthcare workers must be familiar with the medications they prescribe and need to know the medications in their specialty that are associated with high risk of detrimental effects. Drug interactions can lead to serious adverse events or decrease drug efficacy. Prescribing health-care workers should ask patients of any use of over-the-counter medications or dietary supplements because they are frequently under reported and may cause drug interactions. Prescribing the generic name of drugs simplifies the communication among healthcare workers, reducing errors. However, patients need to be educated that their medication may be called by different names (brand and generic name) and they should be encouraged to keep a list of their medications, including both the brand and generic name of each drug.

Packing of surgical instruments

To be effective, the sterilizing agent must come into direct contact with all surfaces of every instrument. Therefore instruments must be packaged (individually or in sets) in such a way to allow adequate exposure to the sterilant, prevent air from being trapped and moisture from being retained during the sterilization process, and ensure sterile transfer to the sterile field.The majority of surgical instruments are made of stainless steel and can be sterilized by steam under pressure. Effective steam sterilization involves the direct contact of all surfaces with steam and the revaporization of water condensate to produce a dry, sterile instrument.

Instrument Packaging

For sterilizing and transporting, instruments are put in a closed container or wrapped individually. Instruments placed in open trays are wrapped as sets. Instruments may be sterilized unwrapped in a high-speed steam sterilizer immediately before use (not recommended), may be prepared in advance as for a case cart, or may be retained in sterile core storage until needed.

Packaging Considerations

The packaging materials for all methods of sterilization should do the following:

Permit penetration of the sterilizing agent to achieve sterilization of all items in the package.

Allow the release of the sterilizing agent at the end of the exposure period and allow adequate drying or aerating.

Withstand the physical conditions of the sterilizing process.

Maintain integrity of the package at varying atmospheric and humidity levels. In dry climates or at high altitudes, some packaging materials are susceptible to rupture during sterilization or dry out and crack in storage.

Provide an impermeable barrier to microorganisms, dust particles, and moisture after sterilization. Items must remain sterile from the time they are removed from the sterilizer until they are used.

Cover items completely and easily and fasten securely with adhesive strip or a heat seal that cannot be resealed after opening. Seal integrity should be tamperproof. A margin of at least 1 inch (2.5 cm) is considered a standard for safety on all sealed packages. Scissors should not be used to cut off the end of a package. Contents cannot be drawn out over this cut end because the contents would be contaminated by the edge of the packaging material. For the same reason, packages are never torn open below a seal.

Resist tears and punctures in handling. If accidental tears and holes do occur, they must be visible.

Identification of the contents and the chemical indicator evidence of exposure to a sterilizing agent should be placed on a label that is visible on the outside of the package.

Be free of toxic ingredients and nonfast dyes.

Be lint-free or low-linting.

Protect the contents from physical damage.

Permit easy removal of the contents with transfer to the sterile field without contamination or delamination (separation into layers).

Be economical.

Packages should be wrapped far enough away from sterile storage areas so that mixing sterile and nonsterile packages is not possible. Cabinets that contain nonsterile items should be labelled conspicuously. A procedure for sending items to and receiving items from the sterilizer should be set up so that sterile and nonsterile packages can never be confused en route. This procedure

must be understood by everyone.

Packages may be wrapped with nonwoven or woven materials sequentially in two layers. Double-thickness wrapping in one layer without using the sequential method is also acceptable. The wrappers can be stitched around the edges or bonded/fused together. Aseptic presentation to the sterile field is the prime consideration. Each facility should determine which method and material is best suited to the clinical environment.

Video link on packaging of surgical instruments

https://www.youtube.com/watch?v=FF5D6flcra0

Theatre layout

The overall floor plan of the surgical suite is divided into three areas or zones, which are directly or indirectly involved with the operative procedure equipment, supplies, and personnel.

For descriptive purposes the zones represent the type of activities, dress code, or restriction for that zone. Each person working within the suite must abide by policies related to the zones.

The Three - Zone Concept

The outer zone - The unrestricted area

Provides an entrance and exit from the surgical suite for personnel, equipment and patients.

Street clothes are permitted in this area, and the area provides access to communicate with personnel within and without the suite.

This zone should contain:

A main access door;

An accessible area for the removal of waste;

A sluice;

Storage for medical and surgical supplies;

An entrance to the changing facilities

The clean or semi-restricted zone

Provides access to the procedure room and peripheral support areas within the surgical unit.

Personnel entering this area must be in proper theatre attire.

Traffic control must be designed to prevent violation of this area by unauthorized personnel or persons improperly attired for this zone.

This zone contains:

The sterile supplies store;

An anesthetic room;

A recovery area (PACU);

A clean corridor;

Rest rooms for the staff.

Aseptic or restricted area

This area should be restricted to the working team.

Staff working in this area should change into theatre clothes, should wear masks and gowns, and, where necessary, should wear sterile gloves

This area includes:

The operating theatre;

A scrub-up area;

The sterile preparation room (preparation of sterile surgical instruments and equipment)

DEPARTMENTAL LAYOUT

Reception area- this the first part of theatre in which the patient come in contact with. The area must create pleasant impression in order to allay anxiety. The area should be large enough in order to allow patient to be transferred from ward stretcher to theatre stretcher Facilities such as telephones and tables for reception staff should be included. Patient are admitted to theatre via this area and checking of various requirement for patient is done.

Holding area-it is a special area inside or adjacent to surgical room. It should be located near reception area. Final identification and assessment before patient is transferred into the operating room. Minor procedures can also be performed i.e inserting catheters, i.v lines. It serves as admission, observation and discharge area.

Scrub room- it is where scrubbing is done to avoid splashing of water which may contaminate the operating room

Operating room- It's a unique acute setting removed from other clinical units. It is controlled geographically, bacteriologically and environmentally. It is restricted in terms of inflow and outflow of personnel. Several methods have been used to prevent transmission of infections i.e Dust collecting surfaces i.e open shelves are made of materials that are resistant to corroding of strong disinfectants. Temperatures are controlled at a range of 20-24 and humidity at 30%-60%. Communication system provides a means of delivery of routine and emergency messages.

5. Anaesthetic room-Should be large enough to accommodate patient trolley, anaesthetic machine and staff movement. Lockable cupboards should be used to accommodate scheduled and controlled drugs, Piped o2, nitrous oxide should be incorporated in the design

Recovery room/ post anaesthetic care unit (PACU)- the goal is to provide total care to postoperative patient. Staff must be vigilant have knowledge on post-operative and post-anaesthesia complications, resuscitation equipment must be available, general lighting, ventilation and must be efficient. Hand washing facilities are essential

Theatre changing room- this is where surgeons nurses and theatre staff change into theatre gowns

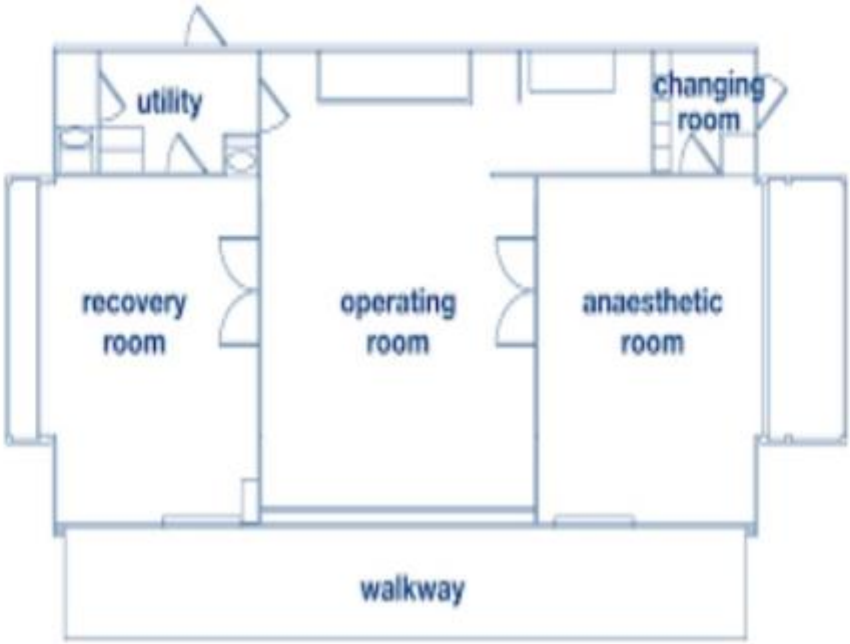
Packing area- instruments are packed according to their specification ready for sterilization

Stores- linens, stationeries and theatre supplies are kept

Utility/sluice room- cleaning of equipment is done and sorting of soiled gowns,

Lounge- located outside operating room. this where surgeons, nursing staff ,anaesthetist and students go to rest after the procedure

Sterilization room(CSSD)



<u>Laminar flow in Operation Theatre</u>

Laminar airflow is a high-pressure, unidirectional air-moving diffuser housed in a cluster in a wall or ceiling enclosure. It is designed to flow uninterrupted from the cleanest area to the less clean area into air return grilles in the lower sidewalls. Cool air from the laminar diffuser travels more quickly than isothermic air. Care is taken to continually monitor the patient's temperature for any hypothermic effect. Hypothermia can support the development of a surgical site infection (SSI).

The airflow is divided into tiny linear columns of cool clean air that generates little velocity and blowing as it courses toward the sterile field. Objects that emit thermal plume, such as spotlights, persons, or equipment, cause the air currents to move in a different direction. *Horizontal airflow* passes from the wall diffuser to an opposite lower return grille. Objects in the air pathway shed particulate into the clean airstream.

*Vertical airflow*styles have fewer obstacles to the line of direction of the air. Vertical downward airflow is designed to flow from the top of the room, over the patient, lights, and equipment, and continue downward forming a canopy toward the lower corners of the room where the air return grilles are located. Studies have shown an 89% reduction in colony-forming units when vertical laminar air flow is used appropriately. This includes making sure that the open instrument tables are well within the confines of the airflow canopy. One additional consideration is when the surgeon and team are directly over the patient to view the surgical site. Particulate from the nonsterile head/cap/mask and neck are passed directly into the sterile field depositing cells into the patient.

Obstacles to Laminar Airflow Effectiveness

Objects that pass between the flow of clean air and the sterile field around the patient shed particles that can contaminate the surgical site. Heat emanating and rising from the team members, the patient, and equipment referred to as thermal plume changes the course of the laminar flow. Several physical barriers and events that interrupt airflow and return include the following:

Spotlights over the field

Ceiling booms with surgical machinery and equipment

Mayo stands over the patient

Microscopes, articulated laser arm, and C-arm

Personnel and patient thermal plume

Opening and closing the OR door (sliding and swinging doors; proportionately higher with swinging doors)

Laminar airflow was first trialed during hip replacement surgery by Sir John Charnley in Great Britain in the 1950s. Charnley believed that if particulate could be removed from the air, the 7% infection rate could drop. His studies showed that the infection rate did fall to less than 2%. Although the laminar system contributes to removing particulates, the improvements in sterile technique and appropriate antibiotic therapy overall may have a larger effect on infection rate. Research has shown that the particulate dispersal is proportionate to the number of personnel in the room, the opening and closing of doors (and their type), placement of instrument tables, and attire worn.

Learning Activities

Case study

The COVID-19 pandemic has infected more than 13 million people worldwide. Among these, an estimated 570,000 healthcare workers have been infected worldwide and approximately 600 in Kenya. Healthcare workers have been the main frontliners in combating the corona virus and therefore require the utmost protection. With the numbers of patients with the virus being on the rise, so are the number of operative cases with coronavirus.

What are the standard precautions used by theater staff to protect themselves from corona virus?

What is the correct sequence for donning PPE?

What is the correct sequence for doffing PPE?

1.2.1.4.2 Field Visit to Operating Theatre

Objectives of the visit

Decontaminate instruments after a surgical procedure

Prepare and pack instruments for steam sterilization

1.2.1.5 Self-Assessment

You are provided with the following questions for self -assessment, attempt them and check your responses

What are the three types of gloves (3) found in theater?

What are three methods of sterilization?

What steps should be followed before assembling instruments for sterilization?

What are characteristics of an ideal packaging material for sterilizing instruments?

What is the purpose of proper waste management?

What are the two main types of laminar flow used in theatre?

What areas are found in the semi-restricted area in theatre?

What are five types of scissors found in theatre?

1.2.1.6 Tools, Equipment, Supplies and Materials

Functional OR

Sterile surgical gloves

Packaging material for sterilization

Decontaminant

Surgical mask

Surgical caps

Theatre scrub gown

Surgical boots

Posters on wearing and removal of PPE

Running water

Hand washing soap

Hand towels

World Health Organization (2020).
https://www.who.int/medical_devices/priority/COVID_19_PPE/en/

RESPONSES

What are the three types of gloves (3) found in theater?

Sterile surgical gloves

Clean gloves

Heavy duty gloves

What are three methods of sterilization?

Thermal

Chemical

Radiation (physical)

What steps should be followed before assembling instruments for sterilization?

Instruments belonging to each set must have passed through terminal cleaning and terminal sterilization before they are safe to handle.

Unless contraindicated, place an absorbent towel or foam in the bottom of the tray to absorb condensate, as for a rigid container with vacuum valves. Include a biologic indicator or chemical integrator in the tray.

Count instruments as they are placed in the tray, and record the number of each type.

Arrange the instruments in a definite pattern to protect them from damage and to facilitate their removal for counting and use. Follow the instrument book or other listing of instruments to be included.

Place heavy instruments, such as retractors, in the bottom of the tray.

Open the hinges and box locks on all hinged instruments.

Place sharp and delicate instruments on top of other instruments. They can be separated with an absorbent material or left in a sterilizing rack with the blades and tips suspended. The blades of scissors, other cutting edges, and delicate tips should not touch other instruments.

Place concave or cupped instruments with the cupped surfaces down so that water condensate does not collect in them during steam sterilization and drying.

Disassemble all detachable parts. Some parts, such as screws and springs, can be put in a peel pouch that is left open. Sealed pouches may not process correctly during the sterilization process.

Separate dissimilar metals. For example, brass knife handles and malleable retractors should be separated from stainless steel instruments. Preferably, put each metal in a separate tray, or separate metals with absorbent material.

Place instruments with a lumen, such as a suction tip, in as near a horizontal position as possible. These instruments should be tilted as little as possible to prevent trapped air or the pooling of water condensate.

Distribute weight as evenly as possible in the tray. Some trays have dividers, clips, and pins that attach to the bottom, which help prevent instruments from shifting and keep them in alignment.

Wrap the tray, or place it in a rigid container. Check woven textile wrappers for holes or abrasion. Sequentially double wrap in a woven or nonwoven material, or use a double thickness wrap in a single-fold configuration.

Place a chemical indicator tab or tape on the outside wrapper or container for proof that the package has been through the parameters necessary for sterilization.

Label the sets appropriately with their intended use (e.g., basic set), the date sterilized, and the process lot control number.

What are characteristics of an ideal packaging material for sterilizing instruments?

Permit penetration of the sterilizing agent to achieve sterilization of all items in the package.

Allow the release of the sterilizing agent at the end of the exposure period and allow adequate drying or aerating.

Withstand the physical conditions of the sterilizing process.

Maintain integrity of the package at varying atmospheric and humidity levels. In dry climates or at high altitudes, some packaging materials are susceptible to rupture during sterilization or dry out and crack in storage.

Provide an impermeable barrier to microorganisms, dust particles, and moisture after sterilization. Items must remain sterile from the time they are removed from the sterilizer until they are used.

Cover items completely and easily and fasten securely with adhesive strip or a heat seal that cannot be resealed after opening.

5. What is the purpose of proper waste management?

Protect people who handle waste items from accidental injury

Prevent the spread of infection to patients, clients, and HCWs

Prevent the spread of infection to the local community

Safely dispose of hazardous materials

6. What are the two main types of laminar flow used in theatre?

Horizontal airflow- passes from the wall diffuser to an opposite lower return grille

Vertical downward airflow - designed to flow from the top of the room, over the patient, lights, and equipment, and continue downward forming a canopy toward the lower corners of the room where the air return grilles are located

7. What areas are found in the semi-restricted area in theatre?

The sterile supplies store;

An anesthetic room;

A recovery area (PACU);

A clean corridor;

Rest rooms for the staff.

8. What are five types of scissors found in theatre?

Sutures scissors-Used to cut sutures and supplies. have blunt points to avoid cutting structures near the suture

Tissue /dissecting scissors-Most have sharp blades .curved or angled blade are needed

mayo scissors- they are curved used to cut heavy tissues

Metzbaum(metz)- used to cut delicate tissues

Wire scissors- have short heavy blades. They are used to cut stainless steel sutures

Dressing/bandage scissors- used to cut drains, dressing.

Learning Outcome 2 Participate in preparation of patient for surgery

Introduction to the learning outcome

This learning outcome specifies the content of competencies required to prepare patient for surgery. It includes:

Identification and preparation of patient for surgery
Patient positions
Prepping and draping the patient
Operation Theatre Ethics
Maintaining Operation Theatre records

Performance Standard

1. Patient is identified according to patient theatre checklist, file and identification band
2. Patient is positioned on operation table according to procedure to be done
3. Patient is prepped according to WHO standards
4. PT records are maintained as per work place policy

Information Sheet

<u>Identification and preparation of patient for surgery</u>

Regardless of the physical setting in which an invasive or surgical procedure will be performed, each patient should be adequately assessed and prepared so that the effects and potential risks of the surgical intervention are minimized. This involves both physical and emotional preparation. Perioperative caregivers should establish a baseline for the patient's preoperative condition so that changes occurring during the perioperative/perianesthesia care period can be easily recognized.

Preoperative Preparation of All Patients

Specific activities such as the preoperative history and physical examination are completed and documented before the patient arrives in the OR. This process can be performed before admission to the hospital or ambulatory care facility; other activities are performed when the patient arrives in the preadmission testing (PAT) area. The preoperative physical preparation is designed to help all patients overcome the stresses of anesthesia, pain, fluid and blood loss, immobilization, and tissue trauma. Preparation often begins before the patient's hospital admission, with the institution of nutritional or drug therapy.

Preadmission Procedures

Some preoperative preparations can be performed in the surgeon's office. Patients are then referred to the preoperative testing center of the hospital or ambulatory care facility. Tests and records should be completed and available before the patient is admitted the day of the surgical procedure. Preadmission tests (PATs) are scheduled according to the guidelines of each facility. Some tests are acceptable only for a 30-day period or are repeated before admission. The type of testing performed depends on the patient’s known or suspected condition and the complexity of the surgical procedure. The preoperative preparations include:

Medical history and physical examination.

Allergies and sensitivities should be noted. The preoperative technician establishes the baseline for the patient’s vital signs

Laboratory tests. Testing should be based on specific clinical indicators or risk factors that could affect surgical management or anesthesia. Tests include age, sex, preexisting disease, magnitude of surgical procedure, and type of anesthesia. Ideally these tests should be completed 24 hours before admission so the results are available for review. Some facilities perform laboratory studies the morning of the procedure.

Hemoglobin, hematocrit, blood urea nitrogen (BUN),and blood glucose may be routinely tested for patients who are age 60 or older.

Hematocrit is usually ordered for women of all ages before the administration of a general anesthetic.

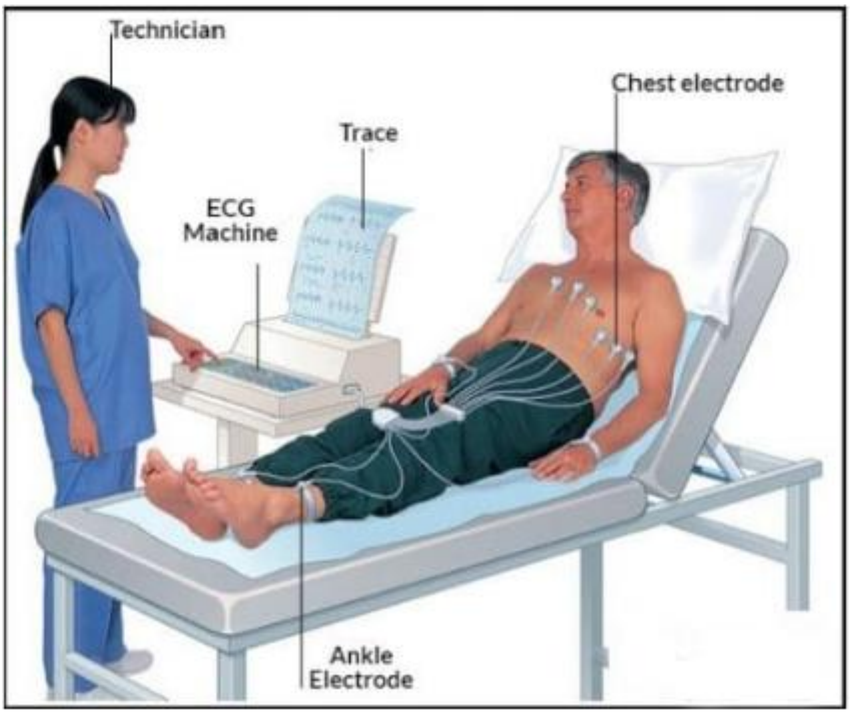
Complete blood count and blood chemistry profile may be indicated. Differential, platelet count, activated partial thromboplastin time, and prothrombin time tests also may be ordered.

Urinalysis may be indicated by the type of surgical procedure, medical history, and/or physical examination.

Blood type and crossmatch. If a transfusion is anticipated, the patient's blood is typed and crossmatched. Many patients prefer to have their own blood drawn and stored for autotransfusion. Patients should be advised that blood banks charge an additional fee to store and preserve autologous blood for personal use. Even if the patient is to have an autotransfusion, his or her blood should still be typed and crossmatched in the event that additional transfusions are needed. If the patient refuses to accept blood transfusions, the appropriate documentation of refusal should be completed according to the policies and procedures of the facility.

Chest x-ray. A preoperative chest x-ray study is not routinely required for all patients. It may be required by facility policy or medically indicated as an adjunct to the clinical evaluation of patients with cardiac or pulmonary disease and for smokers, patients age 60 years or older, and cancer patients.

Electrocardiogram (ECG). If the patient has known or suspected cardiac disease, an ECG is mandatory. Depending on the policy of the facility, an ECG may be routine for patients ages 40 years or older.



Diagnostic procedures. Special diagnostic procedures are performed when specifically indicated (e.g., Doppler studies for vascular surgery).

Written instructions. The patient should receive written preoperative instructions to follow before admission for the surgical procedure. The written literature should be written with the patient's literacy level in mind. Additional methods of instruction should be made available for patients with special needs. Do not use abbreviations or acronyms because they can be confused and misunderstood. Web-based videos can be helpful for patients who cannot read. These instructions should be reviewed with the patient in the surgeon's office or in the preoperative testing center.

GETBETTER HOSPITAL

PREOPERATIVE INSTRUCTIONS

Your doctor has scheduled your surgery to be performed at Getbetter Hospital on ______ (day) ______ (date) at ______ AM/PM. Please arrive at __________ AM/PM.

If you develop a cold, flu, or illness before surgery or cannot keep your surgery appointment, please call your surgeon.

DIETARY RESTRICTIONS

- Avoid alcoholic beverages and cigarette smoking for at least 24 hours before surgery.
- Finish dinner the evening before surgery no later than 8 PM.
- Nothing to eat or drink after midnight the night before surgery. (This includes candy, gum, mints, Lifesavers, ice chips, and water.)
- Eating or drinking may result in the delay or cancellation of surgery.
- Medications to be taken the day of surgery ______________ with sip (1 ounce) of water.
- Medications *not* to be taken the day of surgery ______________________________.

FOR YOUR SAFETY AND COMFORT

- No makeup—it may cause eye irritation or corneal abrasions while under anesthesia.
- Remove nail polish and artificial nails from at least one finger on each hand—oxygen is measured by placing a sensor on your fingertip.
- Do not bring valuables or jewelry—jewelry must be removed before surgery.
- Dress simply in loose-fitting clothes (sweatsuits are ideal).

UPON YOUR ARRIVAL

- Report to Main Lobby Admissions Desk.

AFTER YOUR SURGERY

- No driving, operating of heavy machinery, heavy physical activity, or decision making for 24 hours after receiving an anesthetic.
- If you are returning home on the same day as your surgery, make arrangements for a responsible adult to accompany you. Your operation cannot be performed if a responsible adult is not with you.
- Call your surgeon for postoperative appointments.

If you have any questions, do not hesitate to call the Preadmission Testing Unit at xxx-xxx-xxxx or The Department of Anesthesia at xxx-xxx-xxxx.

Patient signature: ______________________________

RN signature: ______________________________

To prevent regurgitation or emesis and aspiration of gastric contents, the patient should not ingest solid foods before the surgical procedure. These instructions are usually stated as NPO after midnight. Solid foods empty from the stomach after changing to a liquid state, which may take up to 12 hours. Clear fluids may be unrestricted until 2 to 3 hours before the surgical procedure, but only at the discretion of the surgeon or anesthesia provider in selected patients. NPO time usually is reduced for infants, small children, patients with diabetes, and older adults prone to dehydration.

The physician may want the patient to take any essential oral medications that he or she normally takes. These can be taken as prescribed with a minimal fluid intake (a few sips of water) up to 1 hour before the surgical procedure.

The skin should be cleansed to prepare the surgical site. Many surgeons want patients to clean the surgical area with an antimicrobial soap preoperatively or have the patient shower with an antimicrobial sponge, commonly chlorhexidine, the morning of surgery. Patients should be told not to allow the soap to get in the eyes or ears. Chlorhexidine preparations can harm the corneas and tympanic membranes. Patients who will undergo a surgical procedure on the face, ear, or neck are advised to shampoo their hair before admission, because this may not be permitted again for a few days or weeks after the procedure. Nail polish and acrylic nails should be removed to permit observation of and access to the nailbed during the surgical procedure. The patient should be advised to uncover at least one fingernail if the anesthesia provider will use these monitoring devices during the procedure. Either the finger or toe can be used when a digit is desired, but the finger is usually more accessible. Some sensors are adhesive and can be placed on an earlobe or across the bridge of the nose. The nailbed is a vascular area, and the color of the nailbed is one indicator of peripheral oxygenation and circulation. The Oxisensor (optode) of a pulse oximeter may be attached to the nailbed to monitor oxygen saturation and pulse rate.

A finger cuff may be used for continuous blood pressure monitoring. Nail polish or acrylic nails inhibit contact between these devices and the vascular bed.

Jewelry and valuables should be left at home to ensure safekeeping. If electrosurgery will be used, patients should be informed that all metal jewelry, including wedding bands and religious artifacts, should be removed to prevent possible burns. Loss prevention is a consideration as well.

Patients should be given other special instructions about what is expected, such as when to arrive at the surgical facility. A responsible adult should be available to take the patient home if the procedure, medication, or anesthetic agent renders the patient incapable of driving. Family members or significant others should know where to wait and where the patient will be taken after the surgical procedure.

Informed consent. The physician should obtain informed consent from the patient or legal designee. After explaining the surgical procedure and its risks, benefits, and alternatives, the surgeon should document the process and have the patient sign the consent form. This documentation becomes part of the permanent record and accompanies the patient to the OR. Policy and state laws dictate the parameters for ascertaining an informed consent.

BOYDVISION CENTRE CONSENT FOR LASER EYE SURGERY:

I, ____

do hereby consent to Dr. Michael J. Boyd, the attending physician and the staff at BoydVision Centre to perform Laser Eye Surgery:

Lasik ______ PRK ______ EpiLasik ______ PTK ______ Flap Lift ______

Right Eye ______ Left Eye ______ Both Eyes ______

I further authorize Dr. Michael J. Boyd and the staff of BoydVision Centre to perform such additional diagnostic or surgical procedures as may be required or deemed advisable to safeguard life or health during the course of the diagnostic or surgical procedure.

I further consent to the administration of topical or local anaesthetic, or medicines, for the purpose of this diagnostic or surgical procedure.

I further agree that Dr. Michael J. Boyd in his discretion may make use of the assistance of other medical staff and may permit them to order or perform all or part of the diagnostic or surgical procedure.

For all patients, providing consent, the anticipated diagnostic or surgical procedure:
1. has been explained
2. the nature, anticipated effect, risks and alternatives understood
3. is in the best interest of the patient.

It is the obligation of the physician seeking consent to be satisfied that these conditions for valid consent have been met.

Name of Patient ______________ Signature of Patient ______________

Name of Witness ______________ Signature of Witness ______________

Name of Surgeon ______________ Signature of Surgeon ______________

Date ______________

Anesthesia assessment. An anesthesia history and physical assessment are performed before a general or regional anesthetic is administered. The history may be obtained by the surgeon, and/or the patient may be asked to complete a questionnaire for the anesthesia provider in the surgeon’s office or in the preoperative testing center. An interview by an anesthesia provider may be conducted before admission if the patient has a complex medical history, is high risk, or has a high degree of anxiety. All patients should understand the risks of and alternatives to the type of anesthetic to be administered. After discussion with the anesthesia provider, the patient should sign an anesthesia consent form.

Patient positions

Positioning for a surgical procedure is important to the patient’s outcome. Proper positioning facilitates preoperative skin preparation and appropriate draping with sterile drapes. Positioning requires a detailed knowledge of anatomy and physiologic principles and familiarity with the necessary equipment. Safety is a prime consideration. Patient position and skin preparation are determined by the procedure to be performed, with consideration given to the surgeon’s choice of surgical approach and the technique of anesthetic administration. Factors such as age, height, weight, cardiopulmonary status, and preexisting disease condition (e.g., arthritis, allergies) also should be incorporated into the plan of care.

Preoperatively, the patient should be assessed for alterations in skin integrity, for joint mobility, and for the presence of joint or vascular prostheses. The expected outcome is that the patient will not be harmed by positioning, prepping, or draping for the surgical procedure. Efficiency of the patient preparation process can be attained by organizing activities in a logical sequence. The main objectives for any surgical or procedural positioning are as follows:

Optimize surgical-site exposure for the surgeon.

Minimize the risk for adverse physiologic effects.

Facilitate physiologic monitoring by the anesthesia provider.

Promote safety and security for the patient.

Responsibility for Patient Positioning

The selection of the surgical position is made by the surgeon in consultation with the anesthesia provider. Adjustments are made as necessary for the administration and monitoring of anesthetic and for maintenance of the patient's physiologic status. The theatre technologist may be responsible for placing the patient in a surgical position, with guidance from the anesthesia provider and the surgeon. In essence, patient positioning is a shared responsibility among all team members. The anesthesia provider has the final word on positioning when the patient's physiologic status and monitoring are in question. In cases of complex positioning or positioning patients who are obese, the plan of care includes the need for additional help in lifting or positioning. Special devices or positioning aids may be necessary. The weight tolerance of the mechanism and balance of the OR bed is critical. Manufacturer recommendations should be consulted for guidance in selecting the appropriate bed. To avoid questions or confusion, the weight tolerance should be clearly labeled on every OR bed.

Timing of Patient Positioning and Anesthetic Administration

Moving the patient from the transport cart to the OR bed or vice versa requires that both surfaces are securely locked and stable. Someone should be stationed on the far side of the receiving surface to prevent the patient from tumbling off the edge. For any patient under the influence of an anesthetic agent or narcotic medication, personnel should be at the head, foot, and both sides of the patient to prevent dependent parts from sliding off the table. The neck of the patient's gown should be untied to prevent entanglement and choking as the patient moves or is moved from one surface to another. After transfer from the transport cart to the OR bed, the patient is usually supine (face up on the back). Privacy is maintained with a warm cotton blanket, and the thigh strap is positioned in clear sight of the entire team. The patient may be anesthetized in a supine position and then repositioned for the surgical procedure. Some patients are positioned and then anesthetized if their physiologic status requires special care. If patients undergo a procedure in a prone position and with general anesthesia, they are anesthetized and intubated on the transport cart. A minimum of four people is required to place the patient safely in the prone position on the OR bed. Commonly, more personnel are needed for a safe transfer between surfaces when the patient is fully under anesthesia and intubated. Several factors influence the time at which the patient is positioned:

the site of the surgical procedure;

the age and size of the patient;

the technique of anesthetic administration;

whether the patient is conscious and in pain while moving. The patient is not moved, positioned, or prepped until the anesthesia provider indicates it is safe to do so.

Preparations for Positioning

Before the patient is brought into the OR, theatre technologist should do the following:

Review the proposed position by referring to the procedure book and the surgeon's preference in comparison with the scheduled procedure.

Ask the surgeon for assistance if unsure how to position the patient.

Assess for any patient-specific positioning and padding needs.

Check the working parts of the OR bed before bringing the patient into the room.

Assemble and test all table attachments and protective pads anticipated for the surgical procedure and have them immediately available for use.

Review the plan of care for unique needs of the patient.

Safety Measures

Safety measures, including the following, are observed while transferring, moving, and positioning patients:

The patient is properly identified before being transferred to the OR bed, and the surgical site is confirmed according to facility policy. The surgeon is required to initial the correct site.

The patient is assessed for mobility status, which includes determination of the patient's ability to transfer between the transport cart and the OR bed. Do not plan to have patients move themselves toward an affected limb or toward the blinded eye.

The OR bed and transport vehicle are securely locked in position, with the mattress stabilized during transfer to and from the OR bed. Untie the ties of the patient's gown, and take care not to allow the patient's gown or blanket to become lodged between the two surfaces or under the bottom of a moving patient. Velcro strips or other means should be used to maintain the stability of the mattresses of the two surfaces.

Two people should assist an awake patient with the transfer by positioning themselves on each side of the patient's transfer path. The person on the side of the transport cart assists the patient in moving toward the OR bed. The person on the opposite side prevents the patient from falling over the edge of the OR bed.

Adequate assistance in lifting unconscious, anesthetized, obese, or weak patients is necessary to prevent injury. A minimum of four people is recommended, and transfer devices and lifters may be used. The patient is moved on the count of three, with the anesthesia provider giving the signal count. Sliding or pulling the patient may cause dermal abrasion or injury to soft tissues. Dependent limbs can create a counterbalance and cause the patient to fall to the floor. Examination gloves should be worn if the patient is incontinent or offers other risk of exposure to blood and body substances.

Anatomic and Physiologic Considerations

A patient's tolerance of the stresses of the surgical procedure depends greatly on normal functioning of the vital systems. The patient's physical condition is considered, and proper body alignment is important. Criteria are met for physiologic positioning to prevent injury from pressure, crushing, pinching, obstruction, and stretching. Each body system is considered when planning the patient's position for the surgical procedure.

Respiratory Considerations

Unhindered diaphragmatic movement and a patent airway are essential for maintaining respiratory function, preventing hypoxia, and facilitating induction by inhalation anesthesia. Chest excursion is a concern because inspiration expands the chest anteriorly. Some positions limit the amount of mechanical excursion of the chest. Some hypoxia is always present in a horizontal position because the anteroposterior diameter of the ribcage and abdomen decreases.The tidal volume, the functional residual capacity of air moved by a single breath, is reduced by as much as one third when a patient lies down because the diaphragm shifts cephalad. Therefore there should be no constriction around the chest or neck. The patient's arms should be at his or her side, on armboards, or otherwise supported not crossed on the chest, unless absolutely necessary for the procedure. Patients have additional respiratory compromise if they are obese, smoke, or have pulmonary disease.

Circulatory Considerations

Adequate arterial circulation is necessary for maintaining blood pressure, perfusing tissues with oxygen, facilitating venous return, and preventing thrombus formation. Occlusion and pressure on the peripheral blood vessels are avoided. Body support and restraining straps must not be fastened too tightly. Anesthetic agents alter normal body circulatory mechanisms, such as blood pressure. Some drugs cause constriction or dilation of the blood vessels, which is further complicated by positioning.

Peripheral Nerve Considerations

Prolonged pressure on or stretching of the peripheral nerves can result in injuries that range from sensory and motor loss to paralysis and muscle wasting. The extremities, and the body, should be well supported at all times. The most common sites of injury in the upper body are the divisions of the brachial plexus and the ulnar, radial, peroneal, and facial nerves; the axons may be stretched or disrupted. Extreme positions of the head and arm greater than 90 degrees can easily injure the brachial plexus and other superficial nerves. Peripheral nerve injury of the lower body can involve the sciatic, ilioinguinal, and peroneal nerves. If the patient is improperly positioned, the ulnar, radial, and peroneal nerves may be compressed against bone, stirrups, upright retractor posts, or the OR bed.

Musculoskeletal Considerations

A strain on muscle groups results in injury or needless postoperative discomfort. A patient who is anesthetized lacks protective muscle tone. If the head is extended for a prolonged time, the patient may have more pain from the resulting stiff neck than from the surgical wound. Care is taken not to hyperextend a joint, which not only causes postoperative pain but also may contribute to permanent injury to an extremity. Elderly or debilitated patients with osteoporosis or other bone disease may suffer fractures. When turning a patient, always keep the spine in alignment by grasping the shoulder girdle and hip in a logrolling fashion. Do not turn or elevate a patient by grasping only a hip or shoulder and twisting the spine. Proper body alignment is maintained.

Soft Tissue Considerations

Body weight is distributed unevenly when the patient lies on the OR bed. Weight that is concentrated over bony prominences can cause skin pressure ulcers and deep tissue injury. These areas should be protected from constant external pressure against hard surfaces, particularly in patients who are thin or underweight. In addition, tissue that is subjected to prolonged mechanical pressure (e.g., a fold in the skin under an obese or malnourished patient) is not adequately perfused. Wrinkled sheets and the edges of a positioning or other device under the patient can cause pressure on the skin. Foam pads are not adequate to relieve pressure because they compress and do not alternate pressure. Towels and sheet rolls do not relieve pressure because they are unyielding to the patient’s body weight. Gel pads are preferred.

Complications caused by positioning

Hemodynamic instability from orthostatic position

Poor ventilation from thoracic compression

Peripheral nerve injury from compression or stretch

Tissue damage from crush or shearing force

Ischemia of hair-bearing scalp, which causes bald spots

Compartment syndrome

Pressure necrosis

Digit amputation in table bends

Blindness from optic nerve ischemia

Corneal abrasion

Ischemic limbs from arterial occlusion

Venous emboli

Vertebral injury

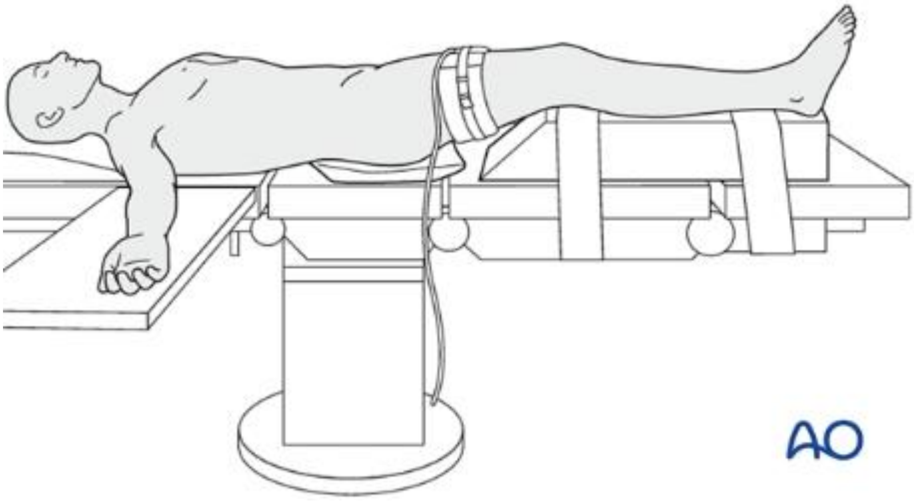
Panic attacks and feelings of claustrophobia in awake patient

Surgical Positions

Supine (Dorsal) Position

The patient lies flat on the back with the arms secured at the sides with the lift sheet; the

palms extend along the side of the body in their natural resting position. The elbows may be protected with padded plastic sleds. The legs are straight and parallel and are in line with the head and spine; the hips are parallel with the spine. A safety belt is placed across the thighs 2 inches above the knees. Small positioning pads may be placed under the head and popliteal area to relieve pressure on the spine as needed. The heels are protected from pressure with a pillow, gel pad, or donut. The feet must not be in prolonged plantar flexion or nerve stretch injury could result. To prevent footdrop, the soles may be supported by a pillow or padded footboard. The supine position is used for procedures on the anterior surface of the body, such as abdominal, abdominothoracic, and some lower extremity procedures.

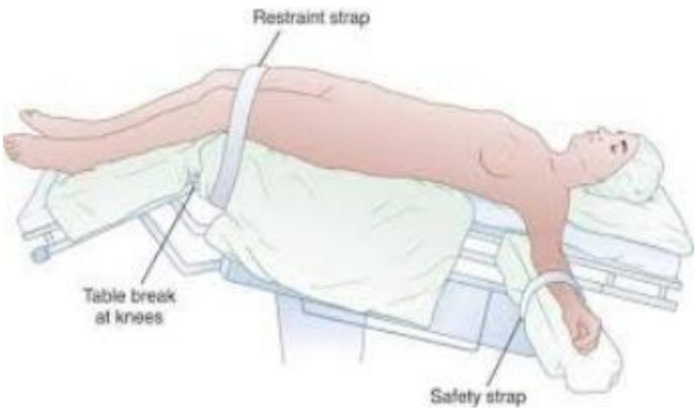


Trendelenburg's Position

With Trendelenburg's position, patients lie on their back in the supine position with the knees over the lower break of the OR bed. The knees must bend with the break of the OR

bed to prevent pressure on the peroneal nerves and veins in the legs. The thigh strap is positioned 2 inches above the knees over the blanket. The entire OR bed is tilted approximately 30 to 45 degrees downward at the head, depending on the surgeon's preference. The foot of the OR bed is lowered to the desired angle to help counterbalance the patient's body weight. Trendelenburg's position is used for procedures in the lower abdomen or pelvis when shifting the abdominal viscera cephalad away from the pelvic area for better exposure. Although surgical accessibility is increased, lung volume is decreased and the heart is mechanically compressed by the pressure of the organs against the diaphragm. Intracranial and intraocular pressure is increased. Therefore the patient remains in this position for as short a time as possible. When returning the patient to a horizontal position, the leg section should be raised first and slowly while venous stasis in

the legs is reversed. The entire OR bed is then leveled.

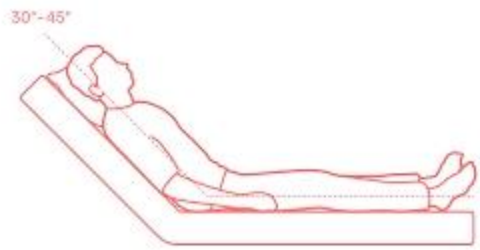


Reverse Trendelenburg's Position

With reverse Trendelenburg's position, patients lie on their back in the supine position. The entire OR bed is tilted 30º to 40º so the head is higher than the feet; a padded footboard is used to prevent the patient from sliding toward the tilt. The thigh safety belt is positioned 2 inches above the knees over the blanket. Small pillows may be placed under the knees. A small pillow or donut is used to stabilize the head. This position is used for thyroidectomy to facilitate breathing and to decrease blood supply to the surgical site (blood pools caudally). It is also used for laparoscopic gallbladder, biliary tract, and stomach procedures to allow the abdominal viscera to fall away from the epigastrium, providing access to the upper abdomen. Venous stasis can cause complications, and prevention of deep vein thrombosis is an important consideration. The use of sequential compression devices, antiembolic stockings, or foot pumps is suggested to improve venous return.

Semi-Fowler's Position

With semi-Fowler's position, the patient is supine with the buttocks at the flex in the OR bed and the knees over the lower break. The foot of the OR bed is lowered slightly, flexing the knees. The body section is raised 45 degrees and thereby becomes the backrest. Arms may rest on armboards parallel to the OR bed or on a large soft pillow on the lap. Care is taken that the arms do not fall dependent from the body during the procedure. The safety belt is secured 2 inches above the knees over the blanket. The entire OR bed is tilted slightly with the head end downward to prevent the patient from slipping toward the foot of the OR bed. Feet should rest on the padded footboard to prevent footdrop. For cranial procedures, the head is supported in a headrest. In this position, the OR bed looks like a modified armchair. This position may be used for shoulder, nasopharyngeal, facial,and breast reconstruction procedures. Complications of this position include air embolus into the venous system, pelvic pooling or venous stasis, hypotension, positional orthopedic injury, and tissue pressure injury and necrosis.



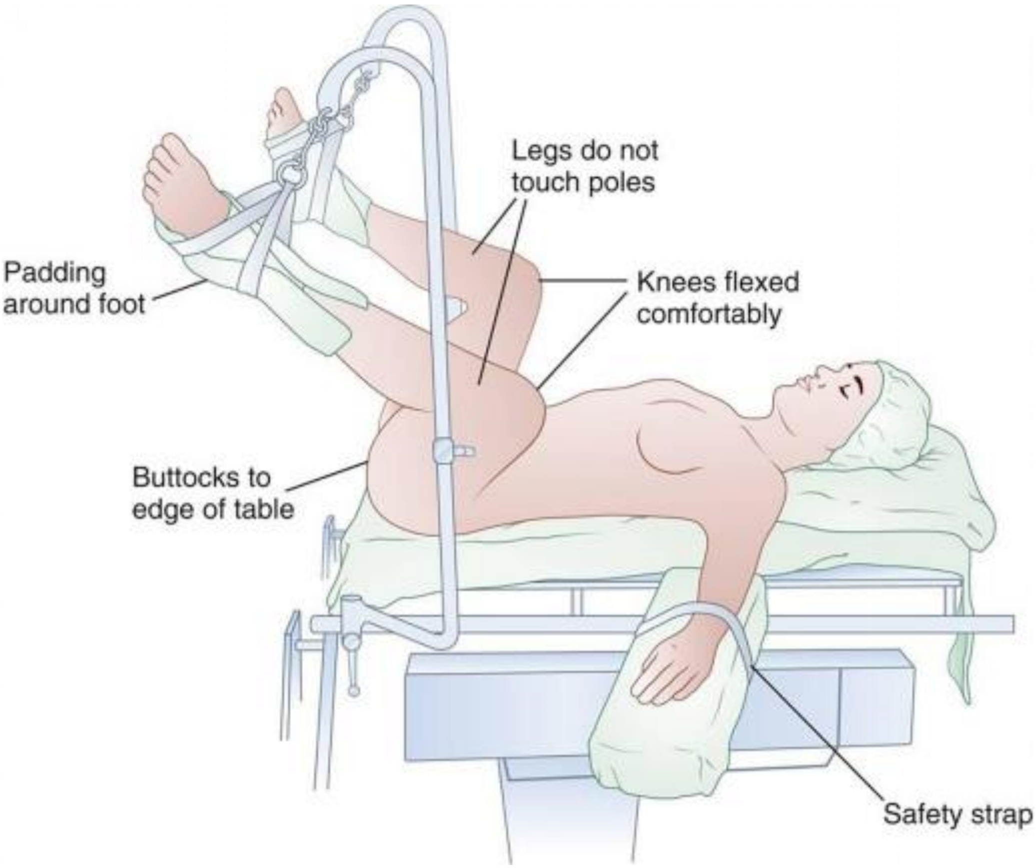
Sitting Position

With the sitting position, the patient is placed in the Fowler’s position except that the torso is completely in an upright position.

Lithotomy Position

The lithotomy position is used for perineal, vaginal, urologic, and rectal procedures. The patient’s buttocks rest along the break between the body and the leg sections of the OR bed. A padded metal footboard is used as an OR bed extension so the patient’s legs do not extend over the foot of the OR bed before the legs are placed in the upright position.

Stirrups are secured in sockets on each side of the OR bed rail at the level of the patient’s upper thighs. They are adjusted at equal height on both sides and at an appropriate height for the length of the patient’s legs to maintain symmetry when the patient is positioned. After the patient is anesthetized, the safety belt is removed and the patient’s legs are raised simultaneously by two people. The safety belt should not remain in place over the abdomen with the patient in the lithotomy position. It causes increased abdominal venous pressure, which in turn increases the risk for blood loss. Each person grasps the sole of a foot in one hand and supports the calf at the knee area with the other. The knees are flexed, and the legs and feet are placed inside the stirrups. Avoid overflexing the hip toward the abdomen to prevent ilioinguinal nerve compression. For sling or candy cane stirrups, the feet are placed in the fabric slings of the stirrups at a 90-degree angle to the abdomen. One padded loop encircles the sole; the other padded loop goes around the ankle.



Prone Position

Prone position is used for all procedures with a dorsal or posterior approach. When the prone position is used, the patient is anesthetized and intubated in the supine position on

the locked transport cart. The patient's arms are along his or her sides. When the anesthesia provider gives permission, the patient is slowly and cautiously shifted toward the OR bed in the supine position and then turned onto the abdomen onto the OR bed. The patient's body is rotated as if rolling a log; a team of at least four to six people is needed to maintain body alignment during this transfer. The anesthesia provider calls the count. The anesthesia provider controls the patient's head and airway while the rest of the patient's body is moved by the team. Chest rolls or bolsters under the axillae and along the sides of the chest from the clavicles to the iliac crests raise the weight of the body from the abdomen and thorax. The weight of the abdomen falls away from the diaphragm and keeps pressure off the vena cava and abdominal aorta. This facilitates respiration,

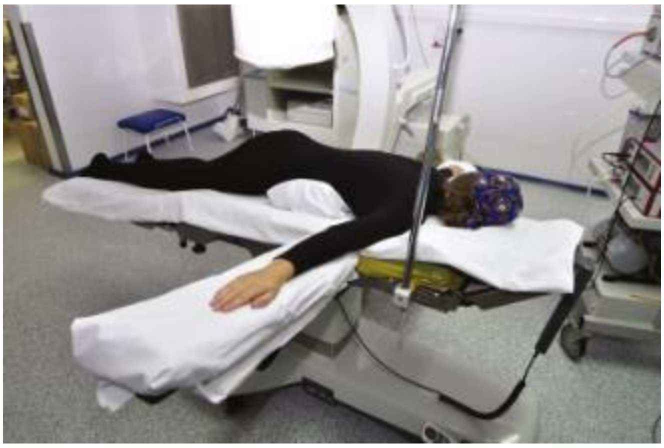
although vital capacity and cardiac index are reduced. To ensure cardiac filling and to reduce hypotension, venous return from the femoral veins and inferior vena cava is uninterrupted.

Female breasts should be moved laterally to reduce pressure on them. Male genitalia should be free from pressure. Pendulous skinfolds should not be crimped under the patient in any manner.

The arms may lie supported along the sides of the body, with the palms up or inward toward the body. An alternate position is to place the arms into a diver's pose by lowering them toward

the floor and rotating (circumducting) them upward in a natural range of motion. Care is taken not to dislocate the shoulders. The armboards are reversed on the table, pointing toward the anesthesia provider. The elbows are padded and are slightly flexed no greater than 90º to prevent overextension, and the palms are down. The arms may extend beyond the head, but not so far as to cause brachial plexus compression or stretch. The head can be turned to one side or positioned face down on a padded headrest to prevent pressure on the ear, eye, and face.

Clearance of the airway must be ensured. A serious complication of the prone position is blindness caused by ischemia of the vascular system of the eye. A pillow or padding under the anterior aspect of the ankles and the dorsa of the feet prevents pressure on the toes and elevates the feet to aid venous return. Do not permit the patient's toes to extend beyond the foot of the bed. Donuts under the knees prevent pressure on the patellae. The safety belt is placed over the calves to prevent flexion of the lower legs. Care is taken not to compress the lower legs. An additional belt can be positioned over the posterior thighs as an added precaution.



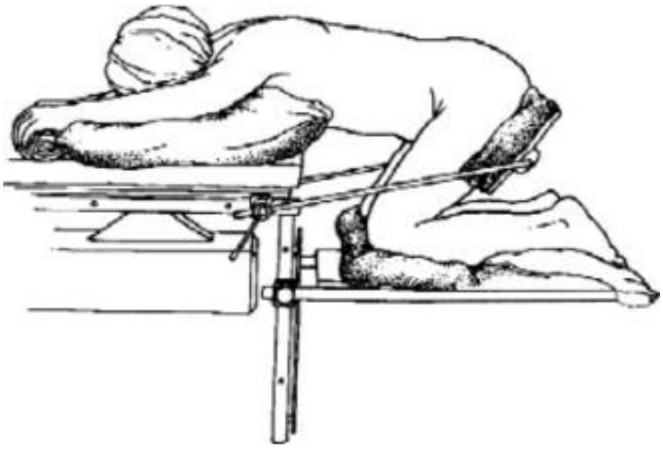
Kraske (Jackknife) Position

With the Kraske position, the patient remains supine until anesthetized and is then turned onto the abdomen (prone position) via rotation. The hips are positioned over the center break of the OR

bed between the body and leg sections. Chest rolls or bolsters are placed to raise the chest if the patient is under general anesthesia. A Kraske pillow can be placed at the level of the anterior thighs to elevate the buttocks. The arms are extended on reverse angled armboards with the elbows flexed and the palms down. The head is to the side and is supported on a donut or pillow. The dorsa of the feet and toes rest on a pillow. The safety belt is placed below the knees. The leg section of the OR bed is lowered the desired amount (usually about 90º), and the entire OR bed is tilted head downward to elevate the hips above the rest of the body. The patient is well balanced on the OR bed . For procedures in the rectal area (e.g., pilonidal sinus, hemorrhoidectomy), the buttocks are retracted with wide tape strips. Because of the dependent position, venous pooling occurs cephalad (toward the head) and caudad (toward the feet). It is important to slowly return the patient to horizontal from this unnatural position to minimize circulatory complications.

Knee-Chest Position

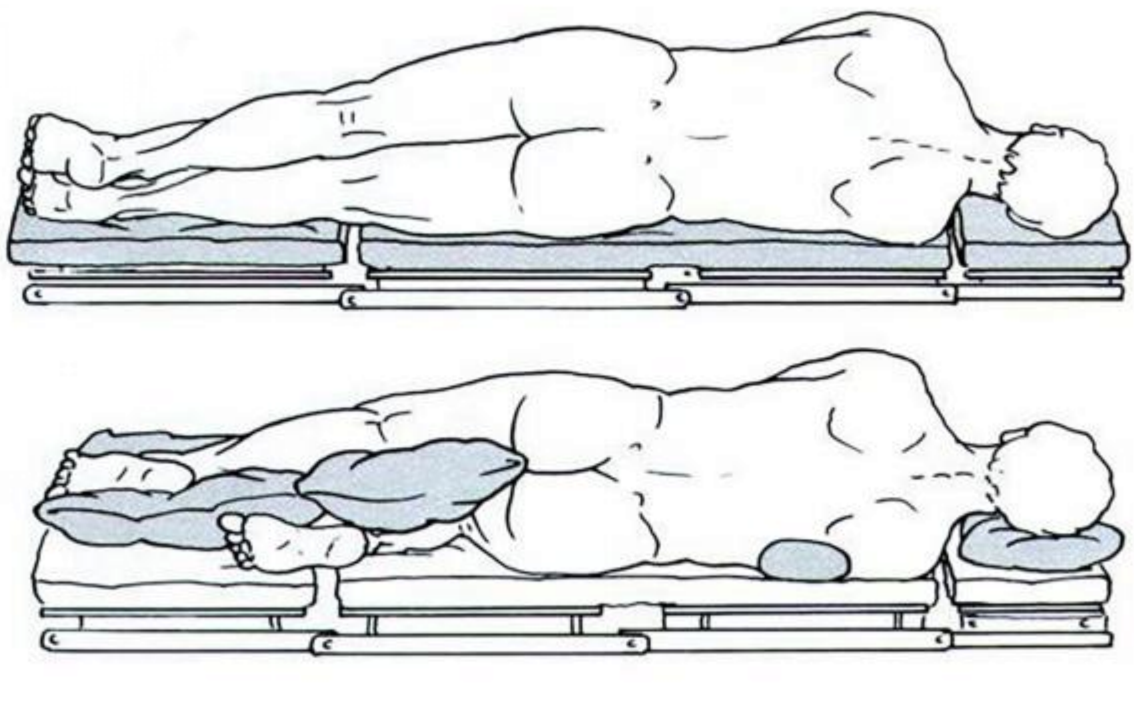
For this position the patient kneels on the lower section; the knees are thus flexed at a right angle to the body. The upper portion of the OR bed may be raised slightly to support the head, which is turned to the side. The arms are placed around the head with the elbows flexed, and a large soft pillow is placed beneath them. The chest rests on the OR bed, and the safety belt is placed above the knees. The entire OR bed is tilted head downward so the hips and pelvis are at the highest point a modified jackknife position.



Lateral Positions

For lateral positioning, the OR bed remains flat. The patient is anesthetized and intubated in the supine position on the OR bed and then turned to the unaffected side. In the right lateral position,

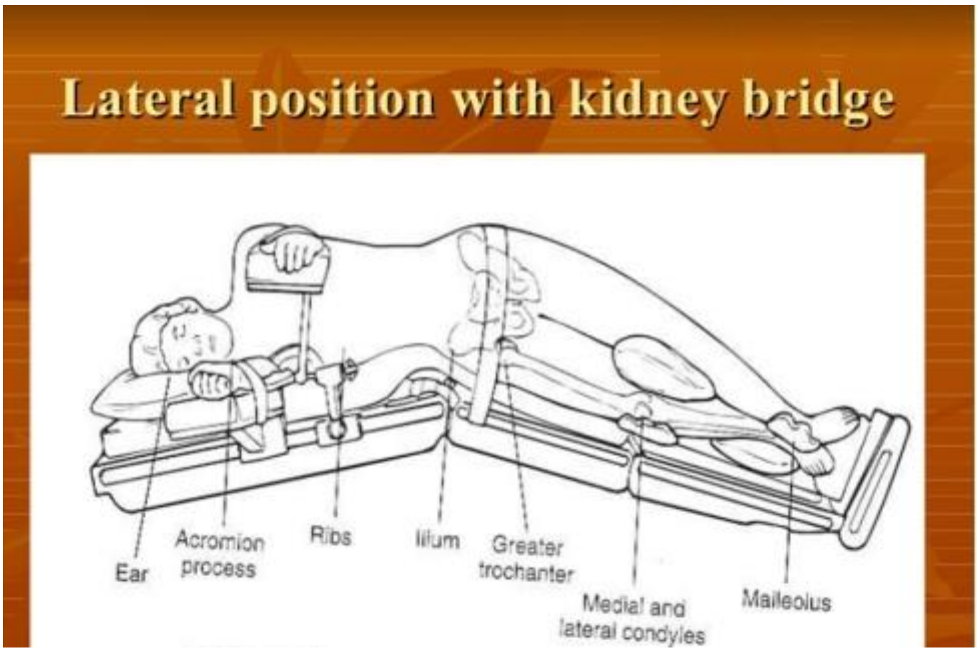
the patient lies on the right side with the left side up (for a left-sided procedure); the left lateral position exposes the right side. The patient is turned by no fewer than four people to maintain body alignment and achieve stability. The patient's back is drawn to the edge of the OR bed. The knee of the lower leg is flexed slightly to provide stabilization, and the upper leg is flexed to provide counterbalance. The flexed knees may require padding to prevent pressure and shearing force. In addition, a large soft pillow is placed lengthwise between the legs to take pressure off the upper hip and lower leg and therefore prevents circulatory complications and pressure on the peroneal nerve. The ankle and foot of the upper leg should be supported to prevent footdrop. Bony prominences are padded. For added stability, a safety belt or a 3-inch-wide tape is placed over the hip. The patient's arms may be placed on a padded double armboard, with the lower arm palm up and the upper arm slightly flexed with the palm down. Blood pressure should be measured from the lower arm. As an alternative, the upper arm can be positioned on a padded Mayo stand. A water bag or pressure reduction pad under the axilla protects neurovascular structures. The shoulders should be in alignment. The patient's head is in cervical alignment with the spine. The head should be supported on a small pillow between the shoulder and neck to prevent stretching the neck and brachial plexus and to maintain a patent airway.



Kidney position

The flank region is positioned over the kidney elevator on the OR bed when the patient is turned onto the unaffected side. The short kidney rest is attached to the body elevator at the patient's back. The longer rest is placed in front at a level beneath the iliac crest to minimize pressure on the abdominal organs. Both rests are well padded. In an obese patient, folds of abdominal tissue may extend over the end of the anterior rest and be bruised if caution is not taken. The OR bed is flexed slightly at the level of the iliac crest so the body elevator can be raised as desired to increase space between the lower ribs and iliac crest. A body strap or wide adhesive tape is placed over the hip to stabilize the patient after the OR bed is flexed and the elevator is raised. The entire OR bed is tilted slightly downward toward the head until the surgical area is horizontal; the upper shoulder and hip should be in a straight line. The upper arm is supported in a double airplane-style armboard. A gel pad is used to support the chest and protect the breasts. An axillary roll is placed to take body weight off the deltoid muscle in the shoulder. Before closure, the OR bed is straightened to allow better approximation of tissues. The term kidney position is used to designate the lateral flexion used to elevate the surgical site. Documentation should clearly state either right or left to indicate which kidney is elevated for the procedure. The kidney position is used for procedures on the kidney and ureter; this position is not well tolerated. Skin and underlying tissue can be damaged by excessive pressure during flexion of the OR bed. OR bed flexion combined with use of the kidney elevator may cause cardiovascular responses. Blood tends to pool in the lower arm and leg. Circulation is further compromised by increased pressure on the abdominal vessels when the kidney elevator is raised. The spine is stressed in a lateral flexed position and

can cause strain on the vertebral structures.



Prepping and draping the patient

Urinary Catheterization

The patient should void to empty the urinary bladder just before transfer to the OR suite unless the procedure requires the bladderto be full, such as for special bladder tone testing procedures. If the patient's bladder is not empty or the surgeon wishes to prevent bladder distention during a long procedure or after the surgical procedure, urinary catheterization may be necessary after the patient is anesthetized. An indwelling Foley catheter may be inserted by a member of the team. This retention catheter maintains bladder decompression to avoid trauma during a lower abdominal or pelvic procedure, to permit accurate measurement of output during or after the surgical procedure, and to facilitate outputand healing after a surgical procedure on genitourinary tract structures. Catheterization is performed after anesthesia is administered and before the patient is positioned for the surgical procedure, except for a patient who will remain in the lithotomy position.The Foley catheter should be inserted before the vaginal or abdominal skin preparation to prevent perineal splash to the surgical site. The vagina can be prepped immediately after the Foley catheter is placed. Gloves should be changed and a new prep set used for the abdominal skin prep.

Principles of Patient Skin Preparation

The purpose of skin preparation is to render the surgical site as free as possible from transient and resident microorganisms, dirt, and skin oil so that the incision can be made through the skin with

minimal danger of infection from this source. Many surgeons prefer to have patients bathe with antimicrobial soap the morning of the surgical procedure.The patient should be advised to avoid the use of body emollients, oils, creams, and lotions after washing. Some products decrease the efficacy of antimicrobial soap, and other products prevent adherence of the return electrode and electrocardiographic (ECG) electrodes to the skin.

The perioperative theatre technologist should assess the patient's skin before, during, and after the prepping process. Documentation of the condition of the patient's skin with notation of lesions or other pertinent markings is important. Abnormal skin irritation, infection, or abrasion on or near the surgical site might be a contraindication to the surgical procedure and is reported to the surgeon. Some patients may have body piercing jewelry that is located on the face, tongue, nose, lip, eyebrow, ears, or other easily identified places. In some cases, the piercing jewelry is located under clothing on the nipples, umbilicus, or genitalia. Body piercing jewelry should be removed for safety reasons before surgery.

Skin Preparation for Specific Anatomic Areas

Head and Neck

Eye

1. The eyebrows are never shaved or removed unless the surgeon deems this essential. Eyebrows do not grow back completely or evenly.

2. The eyelashes may be trimmed, if ordered by the surgeon, with fine iris scissors coated with sterile water-soluble lubricant to catch the lashes.

3. The eyelids and periorbital areas are cleansed with a nonirritating antiseptic agent and then rinsed with warm sterile water. The prep starts centrally and extends to the periphery (i.e., from the center of the lid to the brow and cheek).

4. The conjunctival sac is flushed with a nontoxic agent, such as sterile normal saline solution,with a bulb syringe. Some surgeons use a dilute iodophor solution (not the detergent or alcohol form). The patient’s head is turned slightly to the affected side. The solution is contained with sponges or an absorbent towel. Care must be taken to prevent prep solution from entering the patient’s ears.

Ears, Face, and Nose

1. Skin surfaces should be cleansed at least to the hairline.

2. Cotton applicators are used for cleansing the nostrils and external ear canals.

3. Protect the eyes with a piece of sterile plastic sheeting. If the patient is awake, ask that the eyes be kept closed during the prep. Cotton balls or nonradiopaque gauze should be placed in

the ears to prevent runoff. Cotton balls may leave small fibers in residual cerumen in the ear that can cause irritation.

Neck

1. One sterile towel is folded under the edge of the blanket and gown, which are turned down almost to the nipple line.

2. The area includes the neck laterally to the table line and up to the mandible, tops of the shoulders, and chest almost to the nipple line.

3. For combined head and neck surgical procedures, include the face to the eyes; the shaved areas of the head, ears, and posterior neck; and the area over the shoulders.

Chest and Trunk

Lateral Thoracoabdominal Area

1. The gown is removed. The blanket is turned down well below the lower limit of the area to be prepared. A towel is folded under the edge of the blanket.

2. The arm is held up during the prep.

3. Beginning at the site of incision, the area may include the axilla, chest, and abdomen from the neck to the iliac crest. For a surgical procedure in the region of the kidney, it extends up to the axilla and down to the pubis. The area also extends beyond midlines, anteriorly and posteriorly, and may include the arm to the elbow .

Chest and Breast

1. The anesthesia provider turns the patient’s face toward the unaffected side.

2. One towel is folded under the blanket edge, just above the pubis. Another is placed on the OR bed under the shoulder and side.

3. The arm on the affected side is held up by grasping the hand and raising the shoulder and axilla slightly from the OR bed.

4. The area includes the shoulder, upper arm down to the elbow, axilla, and chest wall to the table line and beyond the sternum to the opposite shoulder . Breast procedures generally require both breasts to be prepped.

Shoulder

1. The anesthesia provider turns the patient’s face toward the opposite side.

2. A towel is placed under the shoulder and axilla.

3. The arm is held up by grasping the hand and elevating the shoulder slightly from the OR bed.

4. The area includes the circumference of the upper arm to below the elbow, from the base of the neck over the shoulder, scapula,and chest to the midline.

Rectoperineal Area

1. With the patient in the lithotomy position, a moisture-proof pad is placed under the buttocks and extends to the kick bucket that receives solutions and discarded prep sponges.

2. The area includes the pubis, external genitalia, perineum and anus, and inner aspects of the thighs Begin the scrub over the pubic area, scrubbing downward over

the genitalia and perineum. Discard the sponge.

4. The inner aspects of the upper third of both thighs are scrubbed with separate sponges, working from groin to distal aspect of thigh.

5. The anus is prepped last.

6. The rectoperineal area is prepped before the abdomen, using a separate prep set and gloves if an abdominal approach is planned.

Vagina

1. Sponge forceps should be included on the preparation table for a vaginal prep because a portion of the prep is done internally. A disposable vaginal prep tray, with sponge sticks included, is

available.

2. With the patient in the lithotomy position, a moisture-proof pad is placed under the buttocks and extends to the kick bucket that receives solutions and discarded prep sponges.

3. A towel is folded above the pubis.

4. Urinary catheterization is performed if indicated. Vaginal and anal flora should not be permitted to enter the sterile environment of the urinary bladder.

5. The external area includes the pubis, vulva, labia, perineum, anus, and adjacent area, including inner aspects of the upper third of the thighs .

6. Begin over the pubic area, scrubbing downward over the vulva and perineum. The inner aspects of the thighs are scrubbed with separate sponges from the labia majora outward.

7. The vagina and cervix are cleansed with sponges on sponge forceps or disposable sponge sticks. The cleansing agent should be applied generously in the vagina because vaginal mucosa has many folds and crevices that are not easily cleansed.

8. After thorough cleansing of the vagina, wipe it out with a dry sponge to prevent the possibility of the fluid entering the peritoneal cavity during the surgical procedure on pelvic organs.

9. The anus is prepped last to prevent intestinal florae from entering the vaginal vault.

Extremities.

1. A moisture-proof pad should be placed on the OR bed under an extremity to retain runoff solution. This is removed after the prep so that the bed is dry under the surgical site. The extremity is supported by personnel wearing sterile gloves and remains elevated until sterile drapes are applied under and around the prepped area.

2. A full circumferential extremity prep may be done in two stages to provide adequate support to joints and to ensure that all areas are scrubbed. It may include the foot for hip, thigh,

knee, and lower leg procedures.

3. Caution must be taken to prevent solution from pooling under a tourniquet. The padding absorbs the solution and could cause tissue maceration. If a nonsterile pneumatic tourniquet is used, it is positioned before the prep and protected with a sterile clear plastic drape. A towel tucked around the tourniquet cuff absorbs excess solution. This is removed before the tourniquet is inflated.

Upper Arm

1. A towel is placed under the shoulder and axilla.

2. The arm is held up by grasping the hand and elevating the shoulder slightly from the OR bed.

3. The area includes the entire circumference of the arm to the wrist, the axilla, and over the shoulder and scapula.

Elbow and Forearm

A towel is placed under the shoulder and axilla.

2. The arm is held up by grasping the hand.

3. The area includes the entire arm from the shoulder and axilla to and including the hand.

Hand

1. Towels may be omitted. The anatomy of the hand furnishes sufficient landmarks to define the area, and towels are apt to slip over the scrubbed area.

2. The arm must be held up by a gloved person supporting it above

the elbow so that the entire circumference can be scrubbed.

3. The area includes the hand and arm to 3 inches (7.5 cm) above

the elbow.

Hip

1. One towel is placed under the thigh on the OR bed. Another towel is placed on the abdomen and folded under the edge of the gown, just above the umbilicus.

2. The leg on the affected side is held up by supporting it just below the knee.

3. The area includes the abdomen on the affected side, the thigh to the knee, and the buttocks to the table line, the groin, and the pubis .

Thigh

1. One towel is placed under the thigh on the OR bed. Another towel is placed on the abdomen and folded under the edge of the gown, just below the umbilicus.

2. The leg is held up by supporting the foot and ankle.

3. The area includes the entire circumference of the thigh and leg to the ankle, over the hip and buttocks to the table line, the groin, and the pubis. Take care not to allow the solution to

pool under the patient's buttocks.

Knee and Lower Leg

1. A towel is placed over the groin.

2. The leg is held up by supporting it at the foot.

3. The area includes the entire circumference of the leg and extends from the foot to the upper part of the thigh.

Ankle and Foot

1. Towels are omitted.

2. The foot is held up by supporting the leg at the knee. A legholder device is useful.

Documentation

Details of the preoperative skin condition and preparation should be documented in the patient's intraoperative record. These should include, but are not limited to, the following:

The condition of the skin around the surgical site and placement sites of the return electrodes

Hair removal, if done, including the method and location, and areas for attachment of monitors or electrodes

Antiseptic solutions, fat solvents, irrigating solutions, and any other agents used

The skin area prepped and skin reaction, if any

The name of the person who did the prep

Draping

Draping is the procedure of covering the patient and surrounding areas with a sterile barrier to create and maintain an adequate sterile field. An effective barrier eliminates or minimizes passage

of microorganisms between nonsterile and sterile areas. Criteria to be met in establishing an effective barrier are that the material must be:

Blood and fluid resistant to keep drapes dry and prevent migration of microorganisms. Material should be impermeable to moist microbial penetration (i.e., resistant to strike-through) and resistant to tearing, puncture, or abrasion that causes fiber breakdown and thus permits microbial penetration.

Lint-free to reduce airborne contaminants and shedding into the surgical site. Cellulose and cotton fibers can cause granulomatous peritonitis or embolize arteries.

Antistatic to eliminate risk of a spark from static electricity.

Sufficiently porous to eliminate heat buildup so as to maintain an isothermic environment appropriate for the patient's body temperature.

Drapable to fit around contours of the patient, furniture, and equipment.

Dull and nonglaring to minimize color distortion from reflected light.

Free of toxic ingredients, such as laundry residues and nonfast dyes.

Draping Materials

Self-Adhering Sheeting. Sterile, waterproof, antistatic, and transparent or translucent plastic sheeting may be applied to dry skin.

Incise Drape. The entire clear plastic drape has an adhesive backing that is applied to skin. This may be applied separately, or the sheeting may be incorporated into the drape sheet. The skin

incision is made through the plastic. Antimicrobial incise drapes have an antimicrobial agent

impregnated in the adhesive or the polymeric film. A film coated with an iodophor-containing adhesive, for example, slowly releases active iodine during the surgical procedure to effectively inhibit proliferation of organisms from the patient's skin. The skin may be prepped with alcohol and allowed to dry before the drape is applied. Time is a factor in microbial accumulation

from resident bacteria. Antimicrobial incise drapes are used to sustain suppression, particularly for procedures that last more than 3 hours.

Towel Drape. A nonwoven towel drape has a band of adhesive along one edge. Used as a draping towel, it remains fixed on skin without towel clips. This is advantageous when clips might

obscure the view of a part exposed to x-rays during the surgical procedure. It also is used to wall off a contaminated area, such as a stoma, from the clean skin area to prevent spilling contents and

causing infection or chemical irritation.

Aperture Drape. Adhesive surrounds a fenestration (opening) in the plastic sheeting. This secures the drape to the skin around the surgical site, such as an eye or ear. Caution is used in applying

this type of drape around the face of a patient who is awake. The patient must have breathing space. If oxygen is used during a local procedure, it can build up under the drape creating a fire hazard

if cautery is used. Some patients experience claustrophobia. Fabric towels may not feel as confining.

Advantages of Self-Adhering Drape Material

Resident microbial flora from skin pores, sebaceous glands, and hair follicles cannot migrate laterally to the incision.

Microorganisms do not penetrate the impermeable material.

Landmarks and skin tones are visible through the transparent plastic.

Inert adhesive holds drapes securely, eliminating the need for towel clips and possible puncture of the patient's skin.

Plastic sheeting conforms to body contours and has elasticity to stretch without breaking its adhesion to skin.

Some self-adhering drapes have sufficient moisture-vapor permeability to reduce excessive moisture buildup that could macerate the skin or loosen adhesive. A nonporous material should not cover more than 10% of the body surface because it may interfere with the patient's thermal regulatory mechanism of perspiration evaporation.

Principles of Draping

The entire team should be familiar with the draping procedure because draping is an important step in the preparation of the patient for a surgical procedure. The scrub person should be knowledgeable and ready to assist with draping. The scrub person is responsible for seeing that necessary articles are arranged in proper sequence on the instrument table. Once a drape is placed

it cannot be moved without contamination. The person responsible for draping the patient may vary, as do materials and styles of drapes used to create a sterile field. The surgeon or assistant usually places the self-adhering incise drape or towels and towel clips to outline the surgical site. The scrub person assists with placing the remainder of the drapes.

During any draping procedure, the circulating nurse should stand by to direct the scrub person as necessary and to watch carefully for breaks in technique. A contaminated drape or exposure

of a nonsterile area is a potential source of infection for the patient.

Basic principles regarding draping

1. Place drapes on a dry area. The area around or under the patient may become damp from solutions used for skin preparation. The circulating nurse removes damp items or covers

the area to provide a dry field on which to lay sterile drapes.

2. Allow sufficient time to permit careful application.

3. Allow sufficient space to observe sterile technique. Do not reach across a nonsterile surface.

4. Handle drapes as little as possible.

5. Never reach across the OR bed to drape the opposite side; go around it.

6. Take towels and towel clips, if used, to the side of the OR bed from which the surgeon is going to apply them before handing them up.

7. Carry folded drapes to the OR bed. Watch the front of the sterile gown; it may bulge and touch the nonsterile OR bed or blanket on the patient. Stand well back from the nonsterile OR bed.

Hold drapes high enough to avoid touching nonsterile areas, but avoid touching the overhead operating light.

Hold a drape high until it is directly over the proper area, and then lay it down where it is to remain. Once a sheet is placed, do not adjust it. Be careful not to slide the sheet out of place when opening the folds.

Protect gloved hands by cuffing the end of the sheet over them. Do not let gloved hands touch the skin of the patient or other nearby items, such as the IV.

8. In unfolding a sheet from the prepped area toward the foot or head of the OR bed, protect the gloved hand by enclosing it in a turned-back cuff of sheet provided for this purpose. Keep

hands at table level.

9. If a drape becomes contaminated, do not handle it further. Discard it without contaminating gloves or other items.

If the end of a sheet falls below waist level, do not handle it further. Drop it, and use another.

If in doubt as to its sterility, consider a drape contaminated.

If a drape is incorrectly placed, discard it. The circulating nurse peels it from the OR bed without contaminating other drapes or the prepped area.

A piercing towel clip that has been fastened through a drape has its points contaminated. Remove it only if absolutely necessary, and then discard it from the sterile setup without touching the points. Cover the area from which it was removed with another piece of sterile draping material.

Operation Theatre Ethics

The ethics of a profession establish the role and scope of professional behavior and the nature of relationships with patients and colleagues.

Autonomy: Self-determination implies freedom of choice and ability to make decisions to determine one's own course of action. Decisions may be made in collaboration with others, based on reasonable and prudent information. Decisions should be acknowledged and respected by others.

Beneficence: Duty to help others seek balance between what is good to do and what might produce harm to another or self.

Nonmaleficence: Duty to do no harm.

Justice: Allocation of human, material, and technologic resources a person has a right to receive or claim (i.e., equality of care).

Veracity: Devotion to truthfulness (i.e., to give accurate information).

Fidelity: Quality of faithfulness, based on trust and honesty, which protects rights of individuals (e.g., dignity, privacy).

Confidentiality: Respect for privileged information received from another person with disclosure only to appropriate others.

An ethical dilemma arises in the work situation when the choice between two or more alternatives creates a conflict between an individual's value system and moral obligation to the patient, to the family or significant others, to the physician, or to the employer and coworkers. Conflicts can be between rights, duties, and responsibilities.

Both legal and ethical considerations can cause conflicts. Legally, a patient has the right to choose among treatment alternatives or the right to refuse treatment. Philosophically, the patient's preference may be different from that of the health care provider. The primary responsibility to the patient is to ensure delivery of safe care. This includes use of appropriate and available technology, but only if this is the patient's choice, with informed consent freely given, or is known to be the patient's wish. Conversely, the patient and the caregiver may be forced to face court-ordered procedures or treatments. This may impose the need to assist in a surgical intervention such as a cesarean section on a woman who has moral or religious objections to this form of treatment but who has been ordered by the court to have the procedure performed for the benefit of the unborn fetus.

Maintaining Operation Theatre records

Clinical records include the admitting diagnosis, the patient's chief complaint, complete history and physical examination findings, the results of laboratory examinations, and the physicians and nurse care plans. Most hospitals are moving toward electronic medical records accessible to all authorized caregivers via computer and the Internet. Records should state the therapy used, including surgical procedures, and include progress notes, consultation remarks, the patient's condition on discharge, and observations in case of death. A summary of the hospitalization experience should be complete. These records are signed in writing or electronically by the physicians and nurses attending the patient.

The anesthesia record completed during the surgical procedure by the anesthesia provider also becomes part of the patient's record. The circulator should document pertinent remarks regarding the care rendered in the OR and the patient's response on the nurses' note sheets or the progress notes. Documentation is a responsibility of all professional team members who implement direct patient care. Notes about the surgical procedure should be explicit, dictated promptly after completion of the procedure, and incorporated into the record of each surgical patient. For the surgeons' convenience, many hospitals have dictating machines or a phone

hookup with the medical records department installed within the OR suite, usually located in the dressing room or lounge. Voice recognition software can be used to dictate procedural notations in the patient's electronic records.

Details of the preoperative and postoperative diagnosis and the surgical procedure itself may have medical and legal significance. Transcription of the surgeon's dictation and maintenance of the patient's chart after discharge from the hospital are responsibilities of the medical records department. The patient's printed chart may be put on microfilm or captured electronically for storage.

Retrieval for review of patient records gives the perioperative theatre technologist the opportunity to evaluate the patient care given. Personnel records can be maintained. Other essential records can be stored in and retrieved from computer databases on a password-controlled basis. The central processing unit for the OR computer system usually is located in or near the control desk. A fax and/or scanner machine may be available for the electronic transfer of documents, records, and patient care orders between the OR and surgeons' offices. Security systems usually can be monitored from the control desk. Alarms are incorporated into electrical and piped-in systems to alert personnel to the location of a system failure. A centralized emergency call system facilitates summoning help. Some facilities have a panic button to summon the security department. Narcotics are kept in locked cabinets and can be signed out only by appropriate personnel.

Access to exchange areas, offices, and storage areas may be limited during evening and night hours and on weekends. Doors may be locked. Some hospitals use alarm systems, video surveillance in hallways and ORs, and electronic metal detection devices to control intruders and to prevent theft or vandalism. Computers and records must be secured to protect patient confidentiality.

1.2.2.4 Learning Activities

1.2.2.4.1 Field Visit to Operating Theatre

Objectives of the visit: Identify equipment for positioning patient during surgery and their functions.

1.2.1.5 Self-Assessment

You are provided with the following questions for self -assessment, attempt them and check your responses

A theatre technologist is preparing a patient for surgery. What are five (5) patient preparation procedures that the technologist will do?

A patient is about to sign an informed consent. What are four (4) important information contained in the consent form?

What is the aim of positioning a patient before a surgical procedure?

During transferring of patients in theater, what are five (5) safety measures that need to be considered?

What are five (5) complications caused by positioning?

Outline characteristics of an effective draping material

What are the ethical principles that need to be adhered to in an operating theater?

What documentation should appear in a patients perioperative records after pre-operative preparation?

1.2.2.6 Tools, Equipment, Supplies and Materials

Functional OR

OR bed

Safety belt

Anesthesia screen

Lift sheet

Arm board

Wrist or arm strap

Upper extremity table

Body rest

Stirrups

footboard

1.2.2.7 References

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Roebuck, A., Harrison, E. (2014). Operating theatre etiquette, sterile technique and surgical site preparation, Surgery 32(3):109–116.

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1.2.2.8 Responses

A theatre technologist is preparing a patient for surgery. What are five (5) patient preparation procedures that the technologist will do?

Taking medical history and physical examination.

Laboratory tests.

Blood type and crossmatch

Diagnostic procedures

Give the patient written preoperative instructions to follow before admission for the surgical procedure

Ensure the patient does not ingest solid foods before the surgical procedure

The skin should be cleansed to prepare the surgical site.

A patient is about to sign an informed consent. What are four (4) important information contained in the consent form?

Details explaining about the surgical procedure

Possible risks involved

Benefits of the surgical procedure

Alternatives to the procedure

What is the aim of positioning a patient before a surgical procedure?

Optimize surgical-site exposure for the surgeon.

Minimize the risk for adverse physiologic effects.

Facilitate physiologic monitoring by the anesthesia provider.

Promote safety and security for the patient.

During transferring of patients in theater, what are five (5) safety measures that need to be considered?

The patient is properly identified before being transferred to the OR bed.

The patient is assessed for mobility status, which includes determination of the patient's ability to transfer between the transport cart and the OR bed.

The OR bed and transport vehicle are securely locked in position, with the mattress stabilized during transfer to and from the OR bed.

Two people should assist an awake patient with the transfer by positioning themselves on each side of the patient's transfer path.

Adequate assistance in lifting unconscious, anesthetized, obese, or weak patients is necessary to prevent injury.

What are five (5) complications caused by positioning?

Hemodynamic instability from orthostatic position

Poor ventilation from thoracic compression

Peripheral nerve injury from compression or stretch

Tissue damage from crush or shearing force

Ischemia of hair-bearing scalp, which causes bald spots

Compartment syndrome

Pressure necrosis

Digit amputation in table bends

Blindness from optic nerve ischemia

Corneal abrasion

Ischemic limbs from arterial occlusion

Venous emboli

Vertebral injury

Panic attacks and feelings of claustrophobia in awake patient

Outline characteristics of an effective draping material

Blood and fluid resistant to keep drapes dry and prevent migration of microorganisms. Material should be impermeable to moist microbial penetration (i.e., resistant to strike-through) and resistant to tearing, puncture, or abrasion that causes fiber breakdown and thus permits microbial penetration.

Lint-free to reduce airborne contaminants and shedding into the surgical site. Antistatic to eliminate risk of a spark from static electricity.

Sufficiently porous to eliminate heat buildup so as to maintain an isothermic environment appropriate for the patient's body temperature.

Drapable to fit around contours of the patient, furniture, and equipment.

Dull and nonglaring to minimize color distortion from reflected light.

Free of toxic ingredients, such as laundry residues and nonfast dyes.

What are the ethical principles that need to be adhered to in an operating theater?

Autonomy

Beneficence

Nonmaleficence

Justice

Veracity

Fidelity

Confidentiality

What documentation should appear in a patients perioperative records after pre-operative preparation?

The condition of the skin around the surgical site and placement sites of the return electrodes

Hair removal, if done, including the method and location, and areas for attachment of monitors or electrodes

Antiseptic solutions, fat solvents, irrigating solutions, and any other agents used

The skin area prepped and skin reaction, if any

The name of the person who did the prep

Learning Outcome 3 Dispose of theatre waste

Introduction to the learning outcome

This learning outcome specifies the content of competencies required to dispose of theatre waste. It includes:

Types of wastes in Operation Theatre

Waste disposal methods

Handling of Operation Theatre waste

Performance Standard

……………xxxxxxxxxxxxxxxxx……………

Information sheet

<u>Types of wastes in Operation Theatre</u>

Infectious waste (e.g., tissues, materials, or equipment that have been in contact with blood or other body fluids).

Pathological waste (e.g., tissues, organs, body parts, blood, infected animals from laboratories, and body fluids).

Sharps (e.g., needles, hypodermic needles, scalpels, and other blades, knives, infusion sets, saws, and broken glass)

Pharmaceutical waste (e.g., expired, spilt, and contaminated pharmaceutical products, discarded bottles or boxes with residues, and drug vials).

Radioactive waste includes solid, liquid, and gaseous materials contaminated with radionuclide.

Genotoxic/cytotoxic waste may include certain cytotoxic drugs often used in cancer therapy, vomit, urine or feces from patients treated with cytotoxic drugs, chemicals, and radioactive material.

Chemical waste consists of discarded solid, liquid, and gaseous chemicals (e.g., from diagnostic and experimental work and from cleaning, housekeeping, and disinfecting procedures).

Non-infectious/general waste includes waste generated from kitchens, packaging material, and from stores.

Waste disposal methods

Method of Disposal of Sharps and Sharps Containers

The main risk that is associated with infection comes from sharps contaminated with blood. Sharps include all objects and materials that pose a potential risk of injury and infection because of their puncture or cutting properties (syringes with needles, blades, wires, broken glass, etc). Place sharps in safety boxes that are resistant to punctures and leakage and are designed so that items can be dropped in using one hand and no item can be removed. The safety box should be marked “Danger Contaminated Sharps” and with the biohazard symbol indicated on the outside of the box. It should be closed when three-quarters full and then placed in a yellow plastic bag or containers with other hazardous health care waste for removal from the procedure area for disposal. The method of choice for destruction of full safety boxes is incineration, preferably in an appropriate double-chamber incinerator. Under exceptional circumstances, full safety boxes may be incinerated in small numbers by open burning. The residues of incineration should be safely buried at a depth greater than 1 meter. Do not, under any circumstances, dispose of used syringes, needles, or safety boxes in normal garbage or dump them without prior treatment.

Encapsulation is recommended as an alternative method to safely dispose of sharps when burning or burying them is not possible. For encapsulation, sharps are collected in puncture-resistant and leak-proof containers.When the container is three-quarters full, a material such as cement, plastic foam, or clay is poured into the container until it is completely filled. After the material has hardened, the container is sealed and buried. It is permissible to encapsulate chemicals or pharmaceutical waste together with sharps.

Disposal of Liquid Contaminated Waste

Liquid contaminated waste requires special handling, because it could pose an infectious risk to HCWs who handle it. Waste water from health care facilities might contain various potentially hazardous components, such as microbiological pathogens, hazardous chemicals, pharmaceuticals, and radioactive isotopes. Wear PPE, including utility gloves, protective eyewear, and a plastic apron when handling and transporting liquid waste. Decontaminate containers by placing them in a 0.5 % chlorine solution for 10 minutes before washing and rinsing them. Remove utility gloves and wash hands and dry hands or use antiseptic hand-rub.

Disposal of solid waste

Disposal of contaminated wastes separately from non-contaminated waste, because contaminated waste needs the following special handling:Wear heavy-duty or utility gloves when handling and transporting solid wastes. Collect the waste containers on a regular basis and transport the burnable ones to the incinerator or area for burning. Remove gloves and wash and dry hands or use an antiseptic hand-rub. Non-contaminated solid wastes should be managed at the health care-facility level or through the local authority disposal system.

Disposal of Hazardous Waste

Hazardous waste material refers to chemical and pharmaceutical waste, as well as any waste that contains heavy metals. Hazardous waste should be incinerated or buried if the quantity is very small. A large quantity of such materials should be sent back to the original supplier.

Pathological waste

Pathological waste includes all organs i.e placenta, tissues, blood and bodily fluids. In operation theatres, all anatomical waste and placentas should be collected separately in a red-colored bin with a red bin liner.Anatomy waste and placentas should be dropped into a concrete-lined placenta pit or buried at a depth greater than 1 meter, inside the health care facility compound in a location totally enclosed and secured from unauthorized access, and at least 100 meters away from any underground water or well. Anatomical waste that cannot be transported and disposed of immediately should be stored in the mortuary.If a patient wants to take home the placenta or body parts for burial first place them in a plastic bag and then into a rigid container for transport.

Infectious Waste

Place infectious waste in a yellow bin with yellow bin liners marked “Biohazard Medical Waste.”When a bag is no more than three-quarters full, seal it with appropriate adhesive tape, remove it, and replace it immediately with a new bag.Incinerate infectious waste in double-chamber incinerators; dispose of ashes from the incinerator in a protected ash pit. An incinerator must attain a temperature of between 800°C and 1,200°C.Sanitary landfill or burial is an alternative solution when underground water is not at risk of contamination.

Hazardous Pharmaceutical Waste and Cytotoxic Waste

Hazardous pharmaceutical waste and cytotoxic waste refers to expired pharmaceuticals or pharmaceuticals that are unusable for other reasons (for example, a call-back campaign). Pharmaceutical waste can be hazardous (cytotoxic). This type of waste should be placed in specific boxes labeled “Danger Hazardous Pharmaceutical and Cytotoxic Waste!” The boxes should be returned to the pharmacy department, which should ensure their disposal at the central level.

Learning Activities

Case study

During routine inspection of the hospital, the Quality control team in your hospital identifies a gap in how the waste is disposed in theater. The team then organizes a continuous medical education to theatre staff on waste management and disposal.

Highlight key information the team will discuss pertaining to waste disposal in the OR?

Discuss waste management in health care setting?

What are the principles of waste disposal in theatre?

Self-Assessment

You are provided with the following questions for self -assessment, attempt them and check your responses

What are examples of contaminated waste in theatre?

Define waste generation

What are four basic principles of waste disposal?

Describe the colour coded bin waste disposal system

An incinerator works best at which temperature?

How is a theatre technologist supposed to handle sharps?

What are five (5) types of waste found in theatre?

Tools, Equipment, Supplies and Materials

Simulated theatre waste

Waste disposal bins

Simulated OR

1.2.2.7 References

Responses

What are examples of contaminated waste in theatre?

Blood

Body fluids

Secretions

Excretions

Define waste generation

Separating contaminated and non-contaminated wastes at the point of generation.

What are four basic principles of waste disposal?

Use appropriate color-coded separate containers for non-infectious, infectious, and highly infectious waste.

Fill the waste containers not more than three-quarters full.

Use color-coded bins and bin liners or label the waste containers

Never sort through contaminated wastes. Do not try to separate non-contaminated waste from contaminated waste, or combustible from non-combustible waste, after they have been combined

Describe the colour coded bin waste disposal system

Non Infectious Waste: Includes food remains, waste paper and packaging material. Disposed in a BLACK color coded bin

Infectious Waste: Includes routine clinical waste e.g. gloves, gauzes and bandages. Disposed in YELLOW color coded bins

Highly Infectious: This includes body tissues/ parts e.g. teeth, placenta. Disposed in RED color coded bins

An incinerator works best at which temperature?

800°C and 1,200°C

How is a theatre technologist supposed to handle sharps during disposal?

Place sharps in safety boxes that are resistant to punctures and leakage and are designed so that items can be dropped in using one hand and no item can be removed.

The safety box should be marked “Danger Contaminated Sharps” and with the biohazard symbol indicated on the outside of the box.

It should be closed when three-quarters full and then placed in a yellow plastic bag or containers with other hazardous health care waste for removal from the procedure area for disposal.

Do not handle sharps unnecessarily.

Always put on heavy-duty gloves when handling sharps waste containers.

In particular, discard all disposable syringes and needles immediately after they are used. The needle should not be recapped or removed from the syringe—the whole combination should be inserted into the safety box directly after use.

What are five (5) types of waste found in theatre?

Infectious waste (e.g., tissues, materials, or equipment that have been in contact with blood or other body fluids).

Pathological waste (e.g., tissues, organs, body parts, blood, infected animals from laboratories, and body fluids).

Sharps (e.g., needles, hypodermic needles, scalpels, and other blades, knives, infusion sets, saws, and broken glass)

Pharmaceutical waste (e.g., expired, spilt, and contaminated pharmaceutical products, discarded bottles or boxes with residues, and drug vials).

Radioactive waste includes solid, liquid, and gaseous materials contaminated with radionuclide.

Genotoxic/cytotoxic waste may include certain cytotoxic drugs often used in cancer therapy, vomit, urine or feces from patients treated with cytotoxic drugs, chemicals, and radioactive material.

Chemical waste consists of discarded solid, liquid, and gaseous chemicals (e.g., from diagnostic and experimental work and from cleaning, housekeeping, and disinfecting procedures).

Non-infectious/general waste includes waste generated from kitchens, packaging material, and from stores.